

# DEEP OPERATOR LEARNING OF MAXIMUM-PRINCIPLE-PRESERVING DISSIPATIVE SEMIGROUPS FOR TIME-DEPENDENT PDES

[AUTHOR NAME]\*

**Abstract.** We formulate operator learning for autonomous evolution PDEs at the semigroup level: rather than learning only a one-step predictor, the learner is a family  $\Phi_\theta(\cdot, \tau)$  intended for repeated composition. We derive discrete stability and error-propagation estimates showing how semigroup defect and trajectory-level approximation error compound under repeated time-stepping. We then construct a semidiscrete (spatially finite-dimensional, temporally continuous) latent dissipative architecture for which invariance of the interior state set, exact semigroup composition, and induced-energy dissipation are exact finite-dimensional guarantees. We prove a transfer theorem that relates the PDE semigroup to the latent architecture through a consistent semidiscrete discretization, and establish universal approximation for gradient-type generators. Numerical experiments on three benchmarks—Fisher–KPP, Allen–Cahn, and viscous Burgers—demonstrate the architectural guarantees, with comparisons against FNO and ResNet baselines showing 14–190× accuracy improvements at short time horizons ( $T = 2$ ) and exact boundary compliance maintained over 200-step repeated compositions where baselines exhibit numerical blowup or growing constraint violations.

**1. Introduction and semigroup formulation.** We study time-dependent PDEs whose solution operators form nonlinear semigroups and satisfy structural laws such as invariant-region preservation and Lyapunov dissipation. These laws are central both to the mathematical interpretation of the dynamics and to the stability of long-time repeated composition. Let  $X$  be a Banach space of states, for example  $L^\infty(\Omega)$ ,  $L^2(\Omega)$ , or a Sobolev space on a spatial domain  $\Omega \subset \mathbb{R}^d$ . Consider an autonomous evolution PDE of the form

$$(1.1) \quad \partial_t u = \mathcal{F}(u), \quad t > 0, \quad u(0) = u_0 \in X.$$

Assume that (1.1) is well-posed [32] and generates a nonlinear solution semigroup

$$(1.2) \quad \mathcal{S}_t : X \rightarrow X, \quad u(t) = \mathcal{S}_t u_0, \quad t \geq 0.$$

Thus

$$(1.3) \quad \mathcal{S}_0 = \text{Id}_X, \quad \mathcal{S}_{t+s} = \mathcal{S}_t \circ \mathcal{S}_s, \quad t, s \geq 0.$$

Our focus is on problems for which the semigroup satisfies two additional structural properties.

*Maximum principle / invariant region..* There exists a closed admissible set  $\mathcal{K} \subset X$  such that

$$(1.4) \quad \mathcal{S}_t(\mathcal{K}) \subset \mathcal{K}, \quad t \geq 0.$$

A typical example is

$$(1.5) \quad \mathcal{K} = \{u \in X : m \leq u(x) \leq M \text{ a.e. in } \Omega\},$$

which encodes a maximum principle, positivity, or boundedness property.

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\*[Affiliation]

*Dissipativity.* There exists a Lyapunov functional

$$(1.6) \quad \mathcal{E} : X \rightarrow \mathbb{R} \cup \{+\infty\}$$

for which

$$(1.7) \quad \mathcal{E}(\mathcal{S}_t u) \leq \mathcal{E}(u), \quad t \geq 0, \quad u \in \mathcal{K}.$$

Depending on the PDE,  $\mathcal{E}$  may be an energy, entropy, free energy, or another dissipated quantity.

Standard operator-learning formulations often emphasize short-time prediction. Our focus is different: we seek a learned evolution family intended for repeated composition in time. The approximation target is therefore the operator family  $(\mathcal{S}_t)_{t \geq 0}$  itself. This emphasis matters because a learned one-step map may fit short-time data and yet accumulate structural errors under repeated composition: small violations of invariance, dissipation, or semigroup consistency can be amplified when the model is used as a time-marching solver. The problem we address is to construct a learned family  $(\Phi_\theta(\cdot, \tau))_{\tau \geq 0}$  that approximates  $(\mathcal{S}_t)_{t \geq 0}$  while retaining (1.4) and (1.7). In other words, the objective is not only local prediction accuracy but a learned evolution family that can be composed in time without destroying the PDE’s structural laws.

*Contributions.* This paper makes three contributions. First, it formulates structure-preserving operator learning at the nonlinear semigroup level: rather than learning an isolated one-step predictor, the target is an evolution family  $\Phi_\theta(\cdot, \tau)$  intended for repeated composition, with the PDE semigroup  $(\mathcal{S}_t)_{t \geq 0}$  as the reference object (Definition 3.1). Second, it derives error-propagation estimates that decompose the two sources of error under repeated composition — trajectory-level approximation and semigroup defect — showing that both enter discrete time-stepping stability in complementary ways (Section 4). Third, it constructs a semidiscrete latent dissipative architecture (Sections 5–6) for which, assuming the latent ODE generates a global flow, invariance of the interior state set  $(m, M)^N$ , exact semigroup composition, and dissipation of an induced discrete energy hold by construction at the finite-dimensional architecture level. A Transfer Theorem (Theorem 8.2) rigorously connects the latent architecture to the target PDE semigroup through a consistent semidiscrete discretization, and universal approximation is established for gradient-type generators (Proposition 9.1). The layered character of the framework is deliberate: the PDE semigroup defines the target specification, while the architectural verification yields structural guarantees that are independent of training quality and hold for every parameter  $\theta$ ; the rigorous transfer between these two levels is formulated as a set of open problems in Section 10.

Section 2 positions the present work relative to the literature. Section 4 develops the discrete stability and error-propagation analysis. Sections 5–6 construct the latent architecture and its neural parametrization. Section 7 presents numerical experiments. Sections 8–9 establish the transfer theorem and universal approximation results.

**2. Relation to existing work.** The present formulation is connected to several lines of work without coinciding with any single one of them. We organize the discussion around five themes.

*Nonlinear semigroup theory.* The mathematical foundation for autonomous evolution PDEs is the theory of nonlinear semigroups [9, 11, 10]. The standard setup associates a closed operator  $\mathcal{F}$  on a Banach space  $X$  with a solution semigroup  $(\mathcal{S}_t)_{t \geq 0}$

satisfying  $\mathcal{S}_0 = \text{Id}$  and  $\mathcal{S}_{t+s} = \mathcal{S}_t \circ \mathcal{S}_s$ . Structural laws such as maximum principles and Lyapunov dissipation are classical features of this framework. Our paper takes these structural properties as the target specification for the learned model, rather than treating them as optional training penalties.

*Structure-preserving numerical methods.* The idea that numerical discretizations should inherit structural properties of the continuous PDE has a long history in numerical analysis. Invariant-region preserving schemes guarantee that the discrete solution remains in a physically admissible set [16, 15, 29]. Strong stability preserving (SSP) time integrators [26] preserve convex constraints and total variation bounds. Geometric numerical integration [25, 24] targets symplectic, volume-preserving, or energy-conserving properties. Implicit-explicit (IMEX) methods [17] separate stiff from non-stiff terms to maintain stability. Our approach differs in that the structural guarantees are built into the architecture of the learned evolution family, rather than into a fixed numerical scheme applied to a known vector field.

*Deep learning for PDEs.* The use of neural networks for PDE problems was catalyzed by the dynamical systems perspective of E [13] and by physics-informed neural networks (PINNs) [12]. Early work on neural-network-based ODE and PDE solvers [33] demonstrated the feasibility of the approach, and high-dimensional PDE solvers based on deep learning [34] extended the scope to previously intractable problems. Deep Galerkin methods [14] and neural-network-based PDE solvers [23] have since expanded the scope of neural approaches. Our work differs from these in target: rather than solving a single PDE instance, we seek to learn the solution operator family itself, as in operator learning approaches.

*Operator learning and neural operators.* DeepONet [6, 30] and the Fourier neural operator (FNO) [5] approximate input-output maps between function spaces. The general neural operator framework [4] provides universality results for such approximations. These approaches typically learn a single-step map; our emphasis is on a whole evolution family intended for repeated composition in time, with structural guarantees that persist under iteration.

*Structure-preserving neural ODEs.* Neural ODEs [1, 20, 35] parametrize continuous-time dynamics with neural networks. Hamiltonian neural networks (HNNs) [2] and SympNets [3] encode symplectic structure. OnsagerNet [7] enforces dissipative structure via the Onsager principle. Port-Hamiltonian neural networks [8] encode dissipative structure through interconnection and damping, and Lyapunov-based neural approaches construct stability certificates for learned dynamics. Symplectic recurrent neural networks [18] and stable neural flows [19] address long-time stability of learned dynamics. Invariant-region-preserving neural networks [21] are the closest in spirit to the present paper. Recent work on the interplay between discretization and regularization in neural ODEs [22] further highlights the importance of understanding how learned dynamics behave under repeated composition. Our contribution adds the semigroup-level formulation and the explicit error-propagation analysis under repeated composition, while the architectural construction (sigmoid logit map + gradient flow with PSD mobility) shares the general principle of encoding constraints via diffeomorphisms to unconstrained spaces.

The distinction of the present paper is that it uses the semigroup viewpoint to keep time composition, invariant-region preservation, and energy decay in the same target specification, while leaving continuous-to-discrete PDE transfer as a separate analytical issue.

**3. Target class of learned semigroups.** We next specify the class of learned families that is admissible for this objective. The basic mathematical object is a parameterized family of learned operators

$$(3.1) \quad \Phi_\theta(\cdot, \tau) : \mathcal{K} \rightarrow \mathcal{K}, \quad \tau \geq 0,$$

intended to approximate the solution operator  $\mathcal{S}_\tau$ . The five requirements below fall into two groups: (3.2)–(3.4) concern approximation of the evolution family itself, whereas (3.5)–(3.6) guarantee that the learned repeated composition stays inside the admissible set and continues to dissipate the chosen Lyapunov functional.

**DEFINITION 3.1** (Structure-preserving deep semigroup learner). *A family  $\Phi_\theta(\cdot, \tau) : \mathcal{K} \rightarrow \mathcal{K}$  is called a structure-preserving deep semigroup learner if it satisfies the following properties.*

(S1) **Identity at zero time:**

$$(3.2) \quad \Phi_\theta(u, 0) = u, \quad u \in \mathcal{K}.$$

(S2) **Generator consistency:** for  $u \in D(\mathcal{F}) \cap \mathcal{K}$ ,

$$(3.3) \quad \Phi_\theta(u, \tau) = u + \tau \mathcal{F}(u) + o(\tau), \quad \tau \downarrow 0.$$

(S3) **Semigroup defect control:** there exists a defect function  $\eta_{\text{sg}} : [0, \infty)^2 \rightarrow [0, \infty)$  such that

$$(3.4) \quad \|\Phi_\theta(\Phi_\theta(u, \tau_1), \tau_2) - \Phi_\theta(u, \tau_1 + \tau_2)\| \leq \eta_{\text{sg}}(\tau_1, \tau_2),$$

for all  $u \in \mathcal{K}$  and all  $\tau_1, \tau_2 \geq 0$ , with  $\eta_{\text{sg}}(\tau_1, \tau_2) \rightarrow 0$  as  $(\tau_1, \tau_2) \rightarrow (0, 0)$ .

(S4) **Maximum-principle preservation:**

$$(3.5) \quad u \in \mathcal{K} \implies \Phi_\theta(u, \tau) \in \mathcal{K}, \quad \tau \geq 0.$$

(S5) **Dissipation preservation:**

$$(3.6) \quad \mathcal{E}(\Phi_\theta(u, \tau)) \leq \mathcal{E}(u), \quad u \in \mathcal{K}, \quad \tau \geq 0.$$

Definition 3.1 isolates the class of learned operator families that are admissible for the present problem. The first three clauses encode approximation of the PDE semigroup, whereas (3.5) and (3.6) encode preservation of the maximum principle and of the dissipation law. In particular, (3.4) measures whether the learned family behaves coherently across different time increments, while (3.5)–(3.6) are the structural conditions that prevent rollout from leaving the physically meaningful regime. This definition therefore serves as the interface between the continuous semigroup problem and the architectural constructions introduced later. It should be read as a PDE-level target specification; the latent architecture constructed later verifies only a finite-dimensional structural subset of these requirements, while generator matching and approximation to the target PDE semigroup remain separate issues.

**REMARK 3.1** (FINITE-DIMENSIONAL ANALOGUE). For the semidiscrete model considered later, the PDE-level Definition 3.1 has a finite-dimensional counterpart. Let  $X_N = \mathbb{R}^N$ ,  $\mathcal{K}_N = [m, M]^N$ , and let  $\mathcal{F}_N : \mathcal{K}_N \rightarrow \mathbb{R}^N$  be a semidiscrete generator. A family  $\Phi_\theta(\cdot, \tau) : \mathcal{K}_N \rightarrow \mathcal{K}_N$  is a *finite-dimensional structure-preserving learner* if it satisfies (3.2) on  $\mathcal{K}_N$ , the finite-dimensional generator relation

$$(3.7) \quad \Phi_\theta(u, \tau) = u + \tau \mathcal{F}_N(u) + o(\tau), \quad u \in \mathcal{K}_N, \quad \tau \downarrow 0,$$

and (3.4)–(3.6) with  $\mathcal{K}$  replaced by  $\mathcal{K}_N$  and  $\mathcal{E}$  replaced by a discrete energy  $\mathcal{E}_N : \mathcal{K}_N \rightarrow \mathbb{R}$ . The latent architecture constructed in Sections 5–6 verifies this finite-dimensional version on the interior set  $(m, M)^N$ , as detailed in Corollary 5.7. The passage from the finite-dimensional learner back to the PDE-level specification is the subject of Problem 10.4 in Section 10.

**4. Repeated composition and discrete stability.** Fix a time step  $\Delta t > 0$ . Once the family  $\Phi_\theta$  is learned, it generates a discrete solver by repeated composition:

$$(4.1) \quad u^{n+1} = \Phi_\theta(u^n, \Delta t), \quad u^0 = u_0,$$

or equivalently

$$(4.2) \quad u^n = (\Phi_\theta(\cdot, \Delta t))^{\circ n}(u_0), \quad n \in \mathbb{N} \cup \{0\}.$$

This section clarifies what is gained by learning the semigroup rather than only a one-step map. Once the model is used for repeated time-stepping, two questions become central: does the discrete orbit remain in the admissible set and dissipate the Lyapunov functional, and how do one-step approximation error and semigroup inconsistency accumulate over many compositions?

**THEOREM 4.1** (Discrete stability inherited from structure). *Assume that (3.5) and (3.6) hold. Then, for every  $u_0 \in \mathcal{K}$ , the discrete orbit (4.2) satisfies*

$$(4.3) \quad u^n \in \mathcal{K}, \quad \mathcal{E}(u^{n+1}) \leq \mathcal{E}(u^n) \leq \mathcal{E}(u^0), \quad n \in \mathbb{N} \cup \{0\}.$$

*In particular, the learned solver preserves the admissible set and satisfies a discrete Lyapunov law.*

*Proof.* The proof is an induction on  $n$ . Since  $u^0 = u_0 \in \mathcal{K}$ , the base step is immediate. If  $u^n \in \mathcal{K}$ , then (3.5) gives  $u^{n+1} = \Phi_\theta(u^n, \Delta t) \in \mathcal{K}$ , and (3.6) yields

$$\mathcal{E}(u^{n+1}) = \mathcal{E}(\Phi_\theta(u^n, \Delta t)) \leq \mathcal{E}(u^n).$$

This proves (4.3).  $\square$

Theorem 4.1 shows that the structural constraints are not merely formal conditions: they immediately turn the learned operator into a stable time-marching scheme. In particular, once (3.5) and (3.6) are built into the model, admissibility and Lyapunov decay propagate automatically to every iterate.

**PROPOSITION 4.2** (Error decomposition through semigroup defect). *Let  $T > 0$  and  $u_0 \in \mathcal{K}$ , and assume that (3.4) and (3.5) hold. Assume also that there exists  $L_T \geq 0$  such that*

$$(4.4) \quad \|\Phi_\theta(v, \tau) - \Phi_\theta(w, \tau)\| \leq (1 + L_T \tau) \|v - w\|, \quad 0 \leq \tau \leq T,$$

*for all  $v, w \in \mathcal{K}$ . Define*

$$(4.5) \quad \varepsilon_{\text{app}}(T; u_0) := \sup_{0 \leq t \leq T} \|\Phi_\theta(u_0, t) - \mathcal{S}_t u_0\|,$$

*and*

$$(4.6) \quad \varepsilon_{\text{sg}}(T, \Delta t) := \sup_{0 \leq t \leq T - \Delta t} \eta_{\text{sg}}(t, \Delta t).$$

*Then, for every integer  $n$  with  $0 \leq n\Delta t \leq T$ ,*

$$(4.7) \quad \left\| (\Phi_\theta(\cdot, \Delta t))^{\circ n}(u_0) - \mathcal{S}_{n\Delta t} u_0 \right\| \leq e^{L_T T} n \varepsilon_{\text{sg}}(T, \Delta t) + \varepsilon_{\text{app}}(T; u_0).$$

*Proof.* Set

$$(4.8) \quad \delta_n := \left\| (\Phi_\theta(\cdot, \Delta t))^{\circ n}(u_0) - \Phi_\theta(u_0, n\Delta t) \right\|.$$

Then  $\delta_0 = 0$ . By (3.5), the iterated state  $(\Phi_\theta(\cdot, \Delta t))^{\circ n}(u_0)$  remains in  $\mathcal{K}$ ; moreover  $\Phi_\theta(u_0, n\Delta t) \in \mathcal{K}$ . Hence the Lipschitz estimate applies to the two arguments below. For  $n\Delta t \leq T - \Delta t$ ,

$$\begin{aligned} \delta_{n+1} &\leq \left\| \Phi_\theta((\Phi_\theta(\cdot, \Delta t))^{\circ n}(u_0), \Delta t) - \Phi_\theta(\Phi_\theta(u_0, n\Delta t), \Delta t) \right\| \\ &\quad + \left\| \Phi_\theta(\Phi_\theta(u_0, n\Delta t), \Delta t) - \Phi_\theta(u_0, (n+1)\Delta t) \right\| \\ &\leq (1 + L_T \Delta t) \delta_n + \varepsilon_{\text{sg}}(T, \Delta t). \end{aligned}$$

A discrete Gronwall argument gives the displayed bound below; we use the coarse estimate  $\sum_{j=0}^{n-1} (1 + L_T \Delta t)^j \leq n e^{L_T T}$ , which is sufficient for the convergence statement.

$$(4.9) \quad \delta_n \leq e^{L_T T} n \varepsilon_{\text{sg}}(T, \Delta t).$$

Combining (4.9) with the triangle inequality yields (4.7).  $\square$

Proposition 4.2 separates two conceptually different error sources. The term  $\varepsilon_{\text{app}}$  measures the trajectory-level semigroup fit from the initial state  $u_0$ , whereas  $\varepsilon_{\text{sg}}$  measures the internal inconsistency of the learned composition law. The estimate shows that long-time repeated composition is controlled not only by short-time approximation quality, but also by how well the learned family behaves as a semigroup. This bound is conditional on the stability hypothesis (4.4) and does not by itself provide a PDE-level approximation theorem.

REMARK 4.1 (DOMAIN OF THE LIPSCHITZ CONDITION). Proposition 4.2 assumes (4.4) holds on the full admissible set  $\mathcal{K}$ . In practice, the Lipschitz verification (Proposition 6.4 below) establishes the condition only on compact subsets  $\mathcal{U} \subset (m, M)^N$ , because  $g^{-1}$  is Lipschitz only on compact subsets of the interior. The error bound (4.7) therefore applies to initial data  $u_0$  whose trajectory under  $\Phi_\theta$  remains in a compact subset of the interior for  $0 \leq t \leq T$ . This restriction is natural: states near the boundary  $\{u_i = m \text{ or } u_i = M\}$  have  $g^{-1}(u_i) \rightarrow \pm\infty$ , and the latent dynamics may degenerate there.

COROLLARY 4.3 (A route to convergence). *Let  $T > 0$  and  $u_0 \in \mathcal{K}$ . Assume that, for a sequence  $\Delta t_m \downarrow 0$  and corresponding learned models satisfying (3.4)–(3.5), one has*

$$(4.10) \quad \varepsilon_{\text{app}}^{(m)}(T; u_0) \rightarrow 0, \quad \frac{\varepsilon_{\text{sg}}^{(m)}(T, \Delta t_m)}{\Delta t_m} \rightarrow 0,$$

and that (4.4) holds with a constant  $L_T$  independent of  $m$ . Then

$$(4.11) \quad \sup_{0 \leq n\Delta t_m \leq T} \left\| (\Phi_\theta^{(m)}(\cdot, \Delta t_m))^{\circ n}(u_0) - \mathcal{S}_{n\Delta t_m} u_0 \right\| \rightarrow 0$$

as  $m \rightarrow \infty$ .

*Proof.* If  $0 \leq n\Delta t_m \leq T$ , then  $n \leq T/\Delta t_m$ . Proposition 4.2 gives

$$\left\| (\Phi_\theta^{(m)}(\cdot, \Delta t_m))^{\circ n}(u_0) - \mathcal{S}_{n\Delta t_m} u_0 \right\| \leq e^{L_T T} T \frac{\varepsilon_{\text{sg}}^{(m)}(T, \Delta t_m)}{\Delta t_m} + \varepsilon_{\text{app}}^{(m)}(T; u_0).$$

Taking the supremum over  $n$  and using (4.10) proves (4.11).  $\square$

**5. A latent structure-preserving architecture.** Corollary 4.3 therefore suggests a concrete design principle: a learned evolution family should control both approximation error and semigroup defect, while preserving the structural laws that guarantee stable repeated composition. The next question is how to build these requirements into an architecture for which the structural clauses of Definition 3.1 hold at the model-class level rather than only through soft penalties during training.

We now work with a semidiscrete finite-dimensional state model (Assumption 5.1 below). The exact statements proved in the remainder of the paper are therefore architecture-level finite-dimensional guarantees, not direct PDE-level transfer results. The rigorous passage from this exact finite-dimensional verification back to the target infinite-dimensional semigroup is deferred to Section 10.

**ASSUMPTION 5.1 (FINITE-DIMENSIONAL STATE MODEL).** *Fix  $N \geq 1$ , and identify the spatially discretized state space with*

$$(5.1) \quad X_N = \mathbb{R}^N.$$

*Assume that the maximum principle is represented by the box constraint*

$$(5.2) \quad \mathcal{K}_N = [m, M]^N = \{u = (u_1, \dots, u_N)^\top \in \mathbb{R}^N : m \leq u_i \leq M, i = 1, \dots, N\}.$$

We regard  $X_N$  and  $\mathcal{K}_N$  as the finite-dimensional counterparts of  $X$  and  $\mathcal{K}$ , respectively. For the inverse-logit latent construction used below, the actual input domain is the interior set  $(m, M)^N$ , because the latent inverse  $g^{-1}$  is only defined there. The resulting learned family is therefore considered as a map

$$(5.3) \quad \Phi_\theta : (m, M)^N \times [0, \infty) \rightarrow \mathcal{K}_N.$$

The construction below follows a simple architectural principle: first lift an interior constrained state into an unconstrained latent variable, then evolve that latent variable by an autonomous dissipative flow, and finally map the latent state back into the admissible set. Boundary states with some component equal to  $m$  or  $M$  are not covered by the present parametrization.

**DEFINITION 5.1 (Bounded latent parametrization).** *Let  $\sigma : \mathbb{R} \rightarrow (0, 1)$  be the sigmoid function*

$$(5.4) \quad \sigma(\xi) = \frac{1}{1 + e^{-\xi}}.$$

*Define  $g : \mathbb{R}^N \rightarrow (m, M)^N$  componentwise by*

$$(5.5) \quad g(z)_i = m + (M - m)\sigma(z_i), \quad i = 1, \dots, N.$$

*Then  $g$  maps the whole latent space into the admissible set. Its inverse on  $(m, M)^N$  is*

$$(5.6) \quad g^{-1}(u)_i = \log\left(\frac{u_i - m}{M - u_i}\right), \quad i = 1, \dots, N,$$

*and its Jacobian is*

$$(5.7) \quad Dg(z) = (M - m) \operatorname{diag}(\sigma(z_i)(1 - \sigma(z_i)))_{i=1}^N.$$

DEFINITION 5.2 (Latent dissipative semigroup). Assume that

$$(5.8) \quad \Psi_\theta \in C^1(\mathbb{R}^N; \mathbb{R}), \quad K_\theta : \mathbb{R}^N \rightarrow \mathbb{R}^{N \times N},$$

and that  $K_\theta(z)$  is symmetric positive semidefinite for every  $z \in \mathbb{R}^N$ . Consider the latent ODE

$$(5.9) \quad \partial_s Z_\theta(s; z_0) = -K_\theta(Z_\theta(s; z_0)) \nabla \Psi_\theta(Z_\theta(s; z_0)), \quad Z_\theta(0; z_0) = z_0.$$

Assume furthermore that this vector field generates a local flow  $\varphi_\theta^\tau$ . The learned operator family is then defined for interior states and for those times  $\tau \geq 0$  for which the latent trajectory through  $g^{-1}(u)$  exists, by

$$(5.10) \quad \Phi_\theta(u, \tau) := g(\varphi_\theta^\tau(g^{-1}(u))).$$

The induced discrete energy on  $(m, M)^N$  is

$$(5.11) \quad \mathcal{E}_\theta(u) := \Psi_\theta(g^{-1}(u)).$$

REMARK 5.1 (REGULARITY HIERARCHY). Definition 5.2 assumes  $\Psi_\theta \in C^1$ , which suffices for the structural properties (Proposition 5.5). The global-flow existence result (Proposition 5.3) and the Lipschitz verification (Proposition 6.4) require the stronger regularity  $\Psi_\theta \in C^2$  and  $K_\theta \in C^1$ , which is ensured by the neural parametrization of Section 6 when the activation  $\rho$  is  $C^2$  (e.g., softplus or tanh).

PROPOSITION 5.3 (Sufficient conditions for global flow). Assume that  $\Psi_\theta \in C^2(\mathbb{R}^N; \mathbb{R})$  satisfies the coercivity condition

$$(5.12) \quad \Psi_\theta(z) \geq c_1 \|z\|^2 - c_2, \quad z \in \mathbb{R}^N,$$

for some constants  $c_1 > 0$  and  $c_2 \geq 0$ . Assume furthermore that  $K_\theta \in C^1(\mathbb{R}^N; \mathbb{R}^{N \times N})$  is symmetric positive semidefinite for every  $z \in \mathbb{R}^N$ . Then the latent ODE (5.9) generates a global flow on  $\mathbb{R}^N$ .

*Proof.* Let  $z_0 \in \mathbb{R}^N$  and let  $Z_\theta(s; z_0)$  denote the maximal solution. Since  $\Psi_\theta \in C^2$  and  $K_\theta \in C^1$ , the vector field  $f_\theta(z) = -K_\theta(z) \nabla \Psi_\theta(z)$  is of class  $C^1$ , hence locally Lipschitz on  $\mathbb{R}^N$ . The energy dissipation identity gives

$$\frac{d}{ds} \Psi_\theta(Z_\theta(s)) = -\nabla \Psi_\theta(Z_\theta(s))^\top K_\theta(Z_\theta(s)) \nabla \Psi_\theta(Z_\theta(s)) \leq 0,$$

so  $\Psi_\theta(Z_\theta(s)) \leq \Psi_\theta(z_0)$  for all  $s$  in the maximal interval of existence. By coercivity (5.12),  $\|Z_\theta(s)\|^2 \leq (\Psi_\theta(z_0) + c_2)/c_1$ , so the solution remains bounded on any finite interval. By the extension theorem for ODEs, the maximal solution exists for all  $s \geq 0$ , i.e., the flow is global.  $\square$

REMARK 5.2 (VERIFIABILITY OF THE GLOBAL-FLOW CONDITIONS). For the neural parametrization of Section 6, condition (5.12) holds if  $\beta_V > 0$  in (6.9), since the quadratic term  $\frac{\beta_V}{2} \|z\|^2$  dominates for large  $\|z\|$ . The regularity  $K_\theta \in C^1$  and positive semidefiniteness hold automatically for the softplus-based mobility parametrization of (6.13). These conditions are therefore verifiable for concrete neural parametrizations and do not constitute additional abstract assumptions.

REMARK 5.3 (EMPIRICAL VALIDITY WITHOUT COERCIVITY). The experiments reported in Section 7 use  $\beta_V = 0$  in (6.9), so the coercivity condition (5.12) is not



satisfied and the global-flow guarantee of Proposition 5.3 does not formally apply. In practice, the learned potential  $V_\theta$  remains bounded on the training distribution, and the RK4 integration with  $M_{\text{sub}} = 30$  substeps on the finite time interval  $[0, T_{\text{max}}]$  does not encounter blowup. However, we do not have a theoretical guarantee of global flow existence for  $\beta_V = 0$ , and the architectural guarantees of Proposition 5.5 (admissibility, exact semigroup, induced-energy dissipation) formally require the global-flow assumption. The experimental results are consistent with the guarantees holding empirically. The global-flow question with  $\beta_V = 0$  is a legitimate analytical concern that we leave as part of Problem 10.1 in Section 10.

PROPOSITION 5.4 (Explicit global-flow criterion for the neural parametrization).

Consider the neural parametrization of Definitions 6.1–6.2. Suppose the following hold:

- (i) The quadratic coefficient satisfies  $\beta_V > 0$  in (6.9).
- (ii) The MLP part  $V_\theta^{\text{mlp}}(\xi) := (c^V)^\top h_{L_V-1}^V(\xi)$  satisfies a growth bound  $|V_\theta^{\text{mlp}}(\xi)| \leq C_1(1 + |\xi|)$  for some  $C_1 > 0$ .
- (iii) The mobility coefficients are uniformly bounded:  $k_{\theta,i}(z) \leq K_{\text{max}}$  for all  $z \in \mathbb{R}^N$  and  $i = 1, \dots, N$ , where  $K_{\text{max}} < \infty$  is a constant depending only on the network parameters.

Then the latent ODE (5.9) generates a global flow on  $\mathbb{R}^N$ .

*Proof.* We verify the hypotheses of Proposition 5.3. Condition (i) ensures  $\Psi_\theta(z) \geq \frac{\beta_V}{2} \|z\|^2 - C_2$  for some  $C_2 \geq 0$ , which is the coercivity condition (5.12) with  $c_1 = \beta_V/2$ . Condition (ii) ensures the MLP component does not overwhelm the quadratic term at large  $\|z\|$ , preserving the coercivity of  $\Psi_\theta$ . The mobility  $K_\theta$  is symmetric positive semidefinite by construction (softplus output). The  $C^1$  regularity of  $K_\theta$  follows from the smooth activation  $\rho = \text{softplus}$ . The result then follows from Proposition 5.3.  $\square$

PROPOSITION 5.5 (Structural properties of the latent-flow construction). Assume that (5.9) generates a global flow on  $\mathbb{R}^N$ . Then, for every  $u \in (m, M)^N$  and every  $\tau, \tau_1, \tau_2 \geq 0$ , the map (5.10) is well defined, takes values in  $(m, M)^N$ , and satisfies

$$(5.13) \quad \Phi_\theta(u, 0) = u,$$

$$(5.14) \quad \Phi_\theta(\Phi_\theta(u, \tau_1), \tau_2) = \Phi_\theta(u, \tau_1 + \tau_2),$$

$$(5.15) \quad \Phi_\theta(u, \tau) \in \mathcal{K}_N,$$

and

$$(5.16) \quad \mathcal{E}_\theta(\Phi_\theta(u, \tau)) \leq \mathcal{E}_\theta(u).$$

*Proof.* Set  $z_0 := g^{-1}(u)$ . Equation (5.13) follows from the identity property of the flow. Equation (5.14) follows from the semigroup property of the autonomous flow  $\varphi_\theta^\tau$ . Since  $g(\mathbb{R}^N) \subset (m, M)^N \subset \mathcal{K}_N$ , the map (5.10) is well defined for all  $\tau \geq 0$ , takes values in  $(m, M)^N$ , and in particular satisfies equation (5.15). Finally,

$$\begin{aligned} \frac{d}{ds} \Psi_\theta(Z_\theta(s; z_0)) &= \nabla \Psi_\theta(Z_\theta(s; z_0))^\top \partial_s Z_\theta(s; z_0) \\ &= -\nabla \Psi_\theta(Z_\theta(s; z_0))^\top K_\theta(Z_\theta(s; z_0)) \nabla \Psi_\theta(Z_\theta(s; z_0)) \\ &\leq 0. \end{aligned}$$

Hence (5.16) follows from (5.11).  $\square$

PROPOSITION 5.6 (Generator expansion and matching condition). *Assume that  $\Psi_\theta \in C^2(\mathbb{R}^N)$  and  $K_\theta \in C^1(\mathbb{R}^N; \mathbb{R}^{N \times N})$ . Then the latent vector field in (5.9) is of class  $C^1$ , so it generates a local  $C^1$  flow. For each fixed  $u \in (m, M)^N$ ,*

$$(5.17) \quad \Phi_\theta(u, \tau) = u - \tau \operatorname{Dg}(g^{-1}(u)) K_\theta(g^{-1}(u)) \nabla \Psi_\theta(g^{-1}(u)) + o(\tau), \quad \tau \downarrow 0.$$

*Consequently, if there exists a semidiscrete generator  $\mathcal{F}_N : (m, M)^N \rightarrow \mathbb{R}^N$  such that the matching condition*

$$(5.18) \quad \operatorname{Dg}(g^{-1}(u)) K_\theta(g^{-1}(u)) \nabla \Psi_\theta(g^{-1}(u)) = -\mathcal{F}_N(u), \quad u \in (m, M)^N,$$

*is satisfied, then*

$$\Phi_\theta(u, \tau) = u + \tau \mathcal{F}_N(u) + o(\tau), \quad \tau \downarrow 0,$$

*which is the finite-dimensional counterpart of (3.3).*

*Proof.* Fix  $u \in (m, M)^N$ , and set  $z_0 := g^{-1}(u)$ . Since the latent vector field

$$f_\theta(z) := -K_\theta(z) \nabla \Psi_\theta(z)$$

is  $C^1$ , the local flow satisfies the standard first-order expansion

$$\varphi_\theta^\tau(z_0) = z_0 + \tau f_\theta(z_0) + o(\tau), \quad \tau \downarrow 0.$$

Because  $g \in C^1(\mathbb{R}^N; (m, M)^N)$ , Taylor expansion at  $z_0$  gives

$$\begin{aligned} \Phi_\theta(u, \tau) &= g(\varphi_\theta^\tau(z_0)) \\ &= g(z_0) + \operatorname{Dg}(z_0)(\varphi_\theta^\tau(z_0) - z_0) + o(\|\varphi_\theta^\tau(z_0) - z_0\|) \\ &= u - \tau \operatorname{Dg}(z_0) K_\theta(z_0) \nabla \Psi_\theta(z_0) + o(\tau). \end{aligned}$$

This is (5.17). If (5.18) holds for some semidiscrete generator  $\mathcal{F}_N$ , then

$$\Phi_\theta(u, \tau) = u + \tau \mathcal{F}_N(u) + o(\tau), \quad \tau \downarrow 0,$$

which is the finite-dimensional counterpart of (3.3).  $\square$

COROLLARY 5.7 (Verification of the finite-dimensional structural clauses). *Assume that the hypotheses of Proposition 5.5 hold. Then the map (5.10) satisfies (3.2) and (3.5) exactly on the interior state set  $(m, M)^N$ , it satisfies (3.4) there with defect function*

$$(5.19) \quad \eta_{\text{sg}}(\tau_1, \tau_2) \equiv 0.$$

*and it satisfies the finite-dimensional dissipation law*

$$(5.20) \quad \mathcal{E}_\theta(\Phi_\theta(u, \tau)) \leq \mathcal{E}_\theta(u), \quad u \in (m, M)^N, \quad \tau \geq 0.$$

*If, in addition, Proposition 5.6 holds and*

$$(5.21) \quad \operatorname{Dg}(g^{-1}(u)) K_\theta(g^{-1}(u)) \nabla \Psi_\theta(g^{-1}(u)) = -\mathcal{F}_N(u), \quad u \in (m, M)^N,$$

*then the finite-dimensional generator relation*

$$(5.22) \quad \Phi_\theta(u, \tau) = u + \tau \mathcal{F}_N(u) + o(\tau), \quad u \in (m, M)^N,$$

*holds as  $\tau \downarrow 0$ .*

Consequently, the latent-flow construction provides an exact finite-dimensional verification of the identity, admissibility, exact semigroup, and induced-energy dissipation properties on the interior state set, while generator matching remains conditional on (5.21) and concerns a semidiscrete generator  $\mathcal{F}_N$ , not yet the original PDE generator  $\mathcal{F}$ .

*Proof.* Equation (3.2) follows from (5.13); equation (3.5) follows from (5.15); (3.4) follows from (5.14), which implies (5.19); and (5.20) follows from (5.16). The final assertion is immediate from Proposition 5.6 and (5.21).  $\square$

**REMARK 5.4 (CONTINUOUS-TIME VS. DISCRETE-TIME SEMIGROUP).** The exact semigroup property (5.19) ( $\eta_{\text{sg}} \equiv 0$ ) holds for the continuous-time latent flow  $\Phi_\theta(u, \tau) = g(\varphi_\theta^\tau(g^{-1}(u)))$ , where  $\varphi_\theta^\tau$  is the exact flow of the latent ODE (5.9). In practice, the latent ODE is integrated by a numerical scheme (e.g., fourth-order Runge–Kutta with  $M_{\text{sub}}$  substeps per  $\tau$ ), which introduces a small integration error. The experiments in Section 7 use  $M_{\text{sub}} = 30$  RK4 substeps, yielding a semigroup defect at the level of the ODE integrator accuracy ( $\sim 10^{-14}$  for 30 RK4 substeps on the smooth latent dynamics). The exact semigroup guarantee is therefore an architecture-level property of the continuous-time formulation; the discrete-time implementation inherits it up to ODE integration accuracy.

**6. Neural parametrization of the latent semigroup.** The previous section identifies the abstract ingredients needed for a structure-preserving semigroup learner. We now turn those ingredients into explicit neural parametrizations of the latent energy, interaction matrix, and mobility.

**DEFINITION 6.1 (Reaction–diffusion latent energy ansatz).** *Let*

$$(6.1) \quad V_\theta \in C^1(\mathbb{R}; \mathbb{R})$$

*be a scalar potential, and let*

$$(6.2) \quad a_{ij}^{(\theta)} = a_{ji}^{(\theta)} \geq 0, \quad i, j = 1, \dots, N.$$

*Define*

$$(6.3) \quad \Psi_\theta(z) = \sum_{i=1}^N V_\theta(z_i) + \frac{1}{2} \sum_{i,j=1}^N a_{ij}^{(\theta)} (z_i - z_j)^2.$$

*If, in addition,*

$$(6.4) \quad K_\theta(z) = \text{diag}(k_{\theta,1}(z), \dots, k_{\theta,N}(z)), \quad k_{\theta,i}(z) \geq 0,$$

*then the latent dynamics takes the componentwise form*

$$(6.5) \quad \partial_s z_i = -k_{\theta,i}(z) \left( V_\theta'(z_i) + 2 \sum_{j=1}^N a_{ij}^{(\theta)} (z_i - z_j) \right), \quad i = 1, \dots, N.$$

**DEFINITION 6.2 (Neural parametrization of latent energy and mobility).** *Fix an activation function  $\rho : \mathbb{R} \rightarrow \mathbb{R}$  of class  $C^1$ , and define*

$$(6.6) \quad \text{softplus}(\xi) = \log(1 + e^\xi).$$

433 For the scalar potential, fix integers  $d_1^V, \dots, d_{L_V-1}^V \in \mathbb{N}$ , together with parameters

$$434 \quad (6.7) \quad W_0^V \in \mathbb{R}^{d_1^V \times 1}, \quad b_0^V \in \mathbb{R}^{d_1^V},$$

435

$$436 \quad W_\ell^V \in \mathbb{R}^{d_{\ell+1}^V \times d_\ell^V}, \quad b_\ell^V \in \mathbb{R}^{d_{\ell+1}^V}, \quad \ell = 1, \dots, L_V - 2, \quad c^V \in \mathbb{R}^{d_{L_V-1}^V}, \quad \beta_V \geq 0.$$

437 Define

$$438 \quad (6.8) \quad h_1^V(\xi) := \rho(W_0^V \xi + b_0^V), \quad h_{\ell+1}^V(\xi) := \rho(W_\ell^V h_\ell^V(\xi) + b_\ell^V), \quad \ell = 1, \dots, L_V - 2,$$

439 and set

$$440 \quad (6.9) \quad V_\theta(\xi) := (c^V)^\top h_{L_V-1}^V(\xi) + \frac{\beta_V}{2} \xi^2.$$

441 For the interaction coefficients, fix an integer  $r_A \geq 1$ , choose symmetric binary  
442 weights  $m_{ij} = m_{ji} \in \{0, 1\}$  and embeddings  $e_i \in \mathbb{R}^{r_A}$  for  $i = 1, \dots, N$ , and define

$$443 \quad (6.10) \quad a_{ij}^{(\theta)} := m_{ij} \text{softplus}(e_i^\top e_j), \quad i, j = 1, \dots, N.$$

444 For the diagonal mobility, fix a stencil radius  $r_K \in \mathbb{N}$  and integers  $d_1^K, \dots, d_{L_K-1}^K \in \mathbb{N}$ .  
445 Fix parameters

$$446 \quad (6.11) \quad W_0^K \in \mathbb{R}^{d_1^K \times (2r_K+1)}, \quad b_0^K \in \mathbb{R}^{d_1^K},$$

447

$$448 \quad W_\ell^K \in \mathbb{R}^{d_{\ell+1}^K \times d_\ell^K}, \quad b_\ell^K \in \mathbb{R}^{d_{\ell+1}^K}, \quad \ell = 1, \dots, L_K - 2, \quad c^K \in \mathbb{R}^{d_{L_K-1}^K}, \quad \beta_K \in \mathbb{R}.$$

449 For each  $z \in \mathbb{R}^N$  and  $i = 1, \dots, N$ , extend boundary values by zero-padding (setting  
450  $z_j = 0$  for indices  $j \notin \{1, \dots, N\}$ ) and define

$$451 \quad (6.12) \quad h_{i,1}^K(z) := \rho(W_0^K(z_{i-r_K}, \dots, z_i, \dots, z_{i+r_K})^\top + b_0^K),$$

452

$$453 \quad h_{i,\ell+1}^K(z) := \rho(W_\ell^K h_{i,\ell}^K(z) + b_\ell^K), \quad \ell = 1, \dots, L_K - 2,$$

454 and set

$$455 \quad (6.13) \quad k_{\theta,i}(z) := \text{softplus}((c^K)^\top h_{i,L_K-1}^K(z) + \beta_K), \quad i = 1, \dots, N.$$

456 Hence the diagonal mobility matrix is

$$457 \quad (6.14) \quad K_\theta(z) := \text{diag}(k_{\theta,1}(z), \dots, k_{\theta,N}(z)).$$

458 The zero-padding convention is used only to make the finite-dimensional stencil  
459 map well defined. It can be replaced by another fixed extension rule, such as peri-  
460 odic, reflective, or boundary-condition-aware padding, without changing the abstract  
461 latent-flow argument above.

462 PROPOSITION 6.3 (Parameterized latent dynamics). Under Definitions 6.1 and  
463 6.2,  $\Psi_\theta \in C^1(\mathbb{R}^N)$  and  $K_\theta(z)$  is symmetric positive semidefinite for every  $z \in \mathbb{R}^N$ .

464 These identities verify the regularity and positivity conditions in Definition 5.2. More-  
 465 over, the coefficients in (6.3) and (6.5) are realized by (6.10) and (6.13). Conse-  
 466 quently,

$$467 \quad (6.15) \quad \Psi_\theta(z) = \sum_{i=1}^N V_\theta(z_i) + \frac{1}{2} \sum_{i,j=1}^N a_{ij}^{(\theta)} (z_i - z_j)^2,$$

468 its gradient satisfies

$$469 \quad (6.16) \quad (\nabla \Psi_\theta(z))_i = (V_\theta)'(z_i) + 2 \sum_{j=1}^N a_{ij}^{(\theta)} (z_i - z_j), \quad i = 1, \dots, N,$$

470 and the latent ODE becomes

$$471 \quad (6.17) \quad \partial_s z_i = -k_{\theta,i}(z) \left( (V_\theta)'(z_i) + 2 \sum_{j=1}^N a_{ij}^{(\theta)} (z_i - z_j) \right),$$

472 for  $i = 1, \dots, N$ .

473 *Proof.* By Definition 6.2, the scalar network  $V_\theta$  is a finite composition of affine  
 474 maps with the  $C^1$  activation  $\rho$ , followed by the smooth quadratic term  $\frac{\beta_V}{2} \xi^2$ ; hence  
 475  $V_\theta \in C^1(\mathbb{R})$ . Likewise, each coefficient  $a_{ij}^{(\theta)} = m_{ij} \text{softplus}(e_i^\top e_j)$  is symmetric and  
 476 nonnegative, and each mobility coefficient  $k_{\theta,i}(z)$  is nonnegative because  $\text{softplus} \geq$   
 477 0. Therefore  $K_\theta(z)$  is diagonal with nonnegative diagonal entries, hence symmetric  
 478 positive semidefinite for every  $z \in \mathbb{R}^N$ .

479 Since  $\Psi_\theta$  in (6.3) is a finite sum of  $C^1$  terms, it belongs to  $C^1(\mathbb{R}^N)$ . Differentiating  
 480 (6.3) with respect to  $z_i$  and using the symmetry  $a_{ij}^{(\theta)} = a_{ji}^{(\theta)}$  yields

$$481 \quad \partial_{z_i} \left( \frac{1}{2} \sum_{p,q=1}^N a_{pq}^{(\theta)} (z_p - z_q)^2 \right) = 2 \sum_{j=1}^N a_{ij}^{(\theta)} (z_i - z_j),$$

482 which gives (6.16). Substituting (6.16) and (6.13) into the latent ODE (5.9) gives  
 483 (6.17). The opening assertions verify the regularity and positivity conditions appear-  
 484 ing in Definition 5.2.  $\square$

485 **PROPOSITION 6.4** (One-step Lipschitz condition for the latent architecture). *As-*  
 486 *sume that the hypotheses of Proposition 5.3 hold and that the latent ODE generates*  
 487 *a global flow. Fix  $T > 0$  and a compact subset  $\mathcal{U} \subset (m, M)^N$ . If  $\Psi_\theta \in C^2(\mathbb{R}^N)$  and*  
 488  *$K_\theta \in C^1(\mathbb{R}^N; \mathbb{R}^{N \times N})$ , then the latent vector field  $f_\theta(z) = -K_\theta(z) \nabla \Psi_\theta(z)$  is locally*  
 489 *Lipschitz on  $\mathbb{R}^N$ , and the flow  $\varphi_\theta^\tau$  is Lipschitz on the preimage  $g^{-1}(\mathcal{U})$  with constant*  
 490 *depending on  $T$  and  $\mathcal{U}$ . Consequently, there exists  $L_T \geq 0$  such that*

$$491 \quad (6.18) \quad \|\Phi_\theta(v, \tau) - \Phi_\theta(w, \tau)\| \leq (1 + L_T \tau) \|v - w\|, \quad v, w \in \mathcal{U}, \quad 0 \leq \tau \leq T.$$

492 *In particular, the hypothesis (4.4) of Proposition 4.2 holds on compact subsets of the*  
 493 *interior state set.*

494 *Proof.* Since  $\Psi_\theta \in C^2$  and  $K_\theta \in C^1$ , the vector field  $f_\theta(z) = -K_\theta(z) \nabla \Psi_\theta(z)$  is  
 495  $C^1$ , hence locally Lipschitz. On the compact set  $g^{-1}(\mathcal{U})$ , let  $L_f$  denote the Lipschitz  
 496 constant of  $f_\theta$  on a compact neighborhood. By the standard Lipschitz dependence of

ODE flows on initial data (obtained via Gronwall's inequality applied to the variational equation), the flow satisfies

$$\|\varphi_\theta^\tau(z_1) - \varphi_\theta^\tau(z_2)\| \leq e^{L_f \tau} \|z_1 - z_2\|, \quad z_1, z_2 \in g^{-1}(\mathcal{U}), \quad 0 \leq \tau \leq T.$$

The sigmoid map  $g$  is Lipschitz on  $\mathbb{R}^N$  with constant  $(M - m)/4$ , and  $g^{-1}$  is Lipschitz on compact subsets of  $(m, M)^N$  with constant  $4/((M - m)\delta)$  where  $\delta = \min_{u \in \mathcal{U}} \min_i \{u_i - m, M - u_i\} > 0$ . Composing the three Lipschitz bounds gives the stated estimate with  $L_T$  depending on  $L_f$ ,  $T$ , and the geometry of  $\mathcal{U}$ .  $\square$

**REMARK 6.1 (TIGHTNESS AT  $\tau = 0$ ).** At  $\tau = 0$ , the composition bound from the proof above gives  $\|\Phi_\theta(v, 0) - \Phi_\theta(w, 0)\| \leq \|v - w\|/\delta$ , which overestimates the true Lipschitz constant of  $\Phi_\theta(\cdot, 0) = \text{Id}$  (which is exactly 1). The tighter form  $\|\Phi_\theta(v, \tau) - \Phi_\theta(w, \tau)\| \leq (1 + L_T \tau) \|v - w\|$  stated in (6.18) follows from the mean value theorem applied to the Jacobian  $D_v \Phi_\theta(v, \tau)$ , which equals  $I$  at  $\tau = 0$  and varies continuously with  $\tau$ . This form preserves the linear-in- $\tau$  growth used in the Gronwall argument of Proposition 4.2.

**7. Benchmark experiments.** We illustrate how the abstract ingredients of Sections 5–6 specialize to three concrete PDE classes: the Fisher–KPP equation, the Allen–Cahn equation, and the viscous Burgers equation. The first two are reaction–diffusion problems with maximum-principle preservation and energy dissipation, while the third tests a stiffer transport-dominated PDE. The rigorous analytical framework (transfer theorem and universal approximation) is developed in Sections 8–9 below; the experiments in this section verify the architectural guarantees empirically before the full theory is presented.

**7.1. Semigroup-based training objectives.** Suppose now that training data contain snapshots of the target evolution, namely triples  $(u, \tau, \mathcal{S}_\tau u)$  with states sampled from the interior set on which the latent construction is defined. The training objective is the one-step prediction loss:

$$(7.1) \quad \mathcal{L}_{\text{step}} = \mathbb{E} \|\Phi_\theta(u, \tau) - \mathcal{S}_\tau u\|^2,$$

*Training strategy in the experiments.* For the latent architecture used below, the structural guarantees (admissibility, exact semigroup, induced-energy dissipation) hold by construction (under the global-flow assumption of Proposition 5.3, which holds empirically for the tested trajectories (the ODE integration does not encounter blowup); see Remark 5.3) for every parameter  $\theta$ . Consequently, the sole training objective is  $\mathcal{L}_{\text{step}}$ . The semigroup property  $\Phi_\theta(\Phi_\theta(u, \tau_1), \tau_2) = \Phi_\theta(u, \tau_1 + \tau_2)$  is enforced architecturally by the ODE flow (uniqueness of solutions), and the RK4 truncation error ( $\sim 10^{-14}$  for 30 substeps) is negligible relative to the approximation error. Energy dissipation is verified as a post-training diagnostic check (not a training loss) to confirm that the architectural guarantee holds empirically.

**7.2. Fisher–KPP equation.** Take the Fisher–KPP equation

$$(7.2) \quad \partial_t u = \nu \Delta u + ru(1 - u), \quad 0 \leq u \leq 1,$$

on a spatial domain  $\Omega \subset \mathbb{R}^d$  with homogeneous Neumann or periodic boundary conditions. The admissible set is  $\mathcal{K} = \{u : 0 \leq u \leq 1\}$  and the dynamics is the  $L^2$ -gradient flow of the energy

$$(7.3) \quad \mathcal{E}(u) = \int_\Omega \left( \frac{\nu}{2} |\nabla u|^2 - r \left( \frac{u^2}{2} - \frac{u^3}{3} \right) \right) dx,$$

so that  $\mathcal{E}(\mathcal{S}_t u) \leq \mathcal{E}(u)$  for all  $t \geq 0$ . The energy  $\mathcal{E}$  is bounded from below on  $\mathcal{K}$  since  $r(u^2/2 - u^3/3) \leq r/6$  for  $u \in [0, 1]$ .

*Spatial discretization and admissible set..* Let  $u = (u_1, \dots, u_N)^\top \in \mathbb{R}^N$  denote the spatially discretized state. The admissible set becomes the box

$$(7.4) \quad \mathcal{K}_N = [0, 1]^N,$$

and the interior set on which the latent construction operates is  $(0, 1)^N$ .

*Latent map..* With  $m = 0$ ,  $M = 1$  the sigmoid parametrization of Definition 5.1 simplifies to

$$(7.5) \quad g(z)_i = \sigma(z_i), \quad g^{-1}(u)_i = \log \frac{u_i}{1 - u_i}, \quad i = 1, \dots, N.$$

*Latent energy and induced discrete energy..* The reaction–diffusion ansatz of Definition 6.1 takes the form

$$(7.6) \quad \Psi_\theta(z) = \sum_{i=1}^N V_\theta(z_i) + \frac{1}{2} \sum_{i,j=1}^N a_{ij}^{(\theta)} (z_i - z_j)^2,$$

where  $V_\theta$  and  $a_{ij}^{(\theta)}$  are parametrized as in Definition 6.2. The induced discrete energy on  $(0, 1)^N$  is

$$(7.7) \quad \mathcal{E}_\theta(u) = \Psi_\theta(g^{-1}(u)).$$

*Architecture guarantees..* By Proposition 5.5, assuming the latent ODE generates a global flow (which holds under the conditions of Proposition 5.3), the learned family  $\Phi_\theta(\cdot, \tau)$  satisfies admissibility ( $\Phi_\theta(u, \tau) \in [0, 1]^N$ ) and induced-energy dissipation  $\mathcal{E}_\theta(\Phi_\theta(u, \tau)) \leq \mathcal{E}_\theta(u)$  exactly on  $(0, 1)^N$ .

*Matching condition..* Let  $\Delta_h$  denote the discrete Laplacian corresponding to the spatial discretization and  $\odot$  denote componentwise multiplication. The generator matching condition (5.18) then reads

$$(7.8) \quad \text{Dg}(g^{-1}(u)) K_\theta(g^{-1}(u)) \nabla \Psi_\theta(g^{-1}(u)) = -(\nu \Delta_h u + r u \odot (1 - u)), \quad u \in (0, 1)^N.$$

This identity is a target to be realized by the neural parametrization of  $\Psi_\theta$  and  $K_\theta$ ; it is not automatically satisfied by the architecture.

**7.3. Allen–Cahn equation.** As a second benchmark, consider the Allen–Cahn equation

$$(7.9) \quad \partial_t u = \varepsilon^2 \Delta u + u(1 - u^2), \quad -1 \leq u \leq 1,$$

on  $\Omega \subset \mathbb{R}^d$  with homogeneous Neumann or periodic boundary conditions. The admissible set is  $\mathcal{K} = \{u : -1 \leq u \leq 1\}$ , and the associated energy is

$$(7.10) \quad \mathcal{E}(u) = \int_\Omega \left( \frac{\varepsilon^2}{2} |\nabla u|^2 + \frac{(1 - u^2)^2}{4} \right) dx,$$

which satisfies  $\mathcal{E}(\mathcal{S}_t u) \leq \mathcal{E}(u)$  for all  $t \geq 0$ . The admissible set becomes  $\mathcal{K}_N = [-1, 1]^N$  with interior  $(-1, 1)^N$ . The latent map, energy ansatz, and matching condition follow the same pattern as for Fisher–KPP with  $m = -1$ ,  $M = 1$ , and the generator matching target

$$(7.11) \quad \text{Dg}(g^{-1}(u)) K_\theta(g^{-1}(u)) \nabla \Psi_\theta(g^{-1}(u)) = -(\varepsilon^2 \Delta_h u + u \odot (1 - u \odot u)).$$

#### 7.4. Experimental setup.

*Architecture details.* Table 1 summarizes the network architectures used in all experiments. The latent learner uses the parametrization of Definitions 6.1–6.2 with a local interaction mask (each site connected to its  $2r_K$  nearest neighbors, periodic padding for  $K_\theta$ , periodic interaction graph for  $a_{ij}^{(\theta)}$ ). The activation function  $\rho$  is softplus throughout, which is  $C^\infty$  and compatible with the regularity requirements of Propositions 5.3–6.4. The ResNet baseline uses a standard residual network with skip connections, trained with a pointwise MSE loss.

TABLE 1  
Architecture details for the latent semigroup learner, FNO, and ResNet baseline.

| Component               | Latent learner                                 | FNO                          | ResNet baseline          |
|-------------------------|------------------------------------------------|------------------------------|--------------------------|
| Potential $V_\theta$    | $L_V=2$ , widths (64, 64), $\beta_V=0^\dagger$ | —                            | —                        |
| Interaction rank $r_A$  | 8 (nearest-neighbor, radius 2)                 | —                            | —                        |
| Mobility $k_{\theta,i}$ | $L_K=2$ , widths (64, 64), $r_K=3$             | —                            | —                        |
| Fourier layers          | —                                              | 4 layers, 16 modes, width 16 | —                        |
| Residual blocks         | —                                              | —                            | 4 blocks, 32 ch.         |
| Total parameters        | ~9.6 K                                         | ~17.5 K                      | ~41.6 K                  |
| Optimizer               | AdamW, wd= $10^{-5}$                           | AdamW, wd= $10^{-5}$         | AdamW, wd= $10^{-5}$     |
| Grad clip               | 1.0                                            | 1.0                          | 1.0                      |
| Learning rate           | $10^{-3}$ (cosine decay)                       | $10^{-3}$ (cosine decay)     | $10^{-3}$ (cosine decay) |
| Batch size              | 64                                             | 64                           | 64                       |
| Training epochs         | 100                                            | 100                          | 100                      |

<sup>†</sup>All experiments use  $\beta_V = 0$ ; the global-flow assumption of Proposition 5.3 is not formally satisfied but holds empirically (see Remark 5.3).

*Training data.* For each benchmark, training data consist of triples  $(u_0, \tau, \mathcal{S}_\tau u_0)$  with 1000 training samples and 50 hold-out test samples. The initial conditions  $u_0$  are drawn from PDE-specific distributions within the interior of the admissible set: for Fisher–KPP,  $u_0 \sim \text{Uniform}[0.3, 0.7]^N$ ; for Allen–Cahn,  $u_0 = \tanh((x-0.5)/(\sqrt{2}\varepsilon)) + \xi$  with  $\xi \sim 0.05(0, I_N)$ ; for Burgers,  $u_0 = \sin(2\pi x/L) \cdot (0.5 + 0.5\eta) + \xi$  with  $\eta \sim \text{Uniform}[0, 1]$  and  $\xi \sim 0.1(0, I_N)$ . The time step is  $\tau = \Delta t = 0.1$  for Fisher–KPP and Allen–Cahn, and  $\tau = \Delta t = 0.05$  for the stiffer Burgers benchmark. The reference solution  $\mathcal{S}_\tau u_0$  is computed by a high-resolution numerical integrator with a finer internal time step  $\Delta t_{\text{ref}}$ : central finite differences with RK4 ( $\Delta t_{\text{ref}} = 0.005$ ) for Fisher–KPP, upwind finite differences with RK4 ( $\Delta t_{\text{ref}} = 0.001$ ) for Burgers, and a semi-implicit IMEX scheme with spectral treatment of the diffusion term ( $\Delta t_{\text{ref}} = 0.0005$ ) for Allen–Cahn.

*Computational cost.* All experiments were conducted on a single NVIDIA GeForce RTX 5080 GPU. The latent learner requires  $M_{\text{sub}} = 30$  RK4 substeps per time step, corresponding to 120 function evaluations per step (vs. 1 for FNO and ResNet). Despite this overhead, the latent learner’s total training time is comparable to the baselines due to its smaller parameter count (9.6 K vs. 17.5 K for FNO and 41.6 K for ResNet). The per-epoch training times are approximately 45 s for the latent learner, 30 s for FNO, and 35 s for ResNet on the Fisher–KPP benchmark ( $N = 64$ , batch size 64).

*Reproducibility.* All experiments use the PyTorch default random initialization. The weight initialization for the latent architecture uses  $\mathcal{N}(0, 0.001)$  for linear layer weights and zero biases, chosen for ODE stability at initialization. Random seeds are not fixed across runs; the reported means and standard deviations are computed



over 50 (Fisher–KPP) or 20 (long-time rollout) independent validation trajectories, providing statistical robustness despite stochastic training.

## 7.5. Results.

*Main comparison (Fisher–KPP).* Table 2 reports the evaluation metrics on 50 hold-out initial conditions, each rolled out over 20 time steps ( $T_{\max} = 2.0$ ).

TABLE 2

*Comparison on the Fisher–KPP benchmark (50 test samples, mean  $\pm$  std). The latent learner’s boundary violation, semigroup defect, and energy monotonicity values confirm the architecture-level guarantees of Propositions 5.5–5.3.*

| Metric              | Latent semigroup                 | FNO                              | ResNet baseline                   |
|---------------------|----------------------------------|----------------------------------|-----------------------------------|
| Parameters          | 9.6 K                            | 17.5 K                           | 41.6 K                            |
| Rollout MSE (final) | $(1.49 \pm 0.15) \times 10^{-4}$ | $(2.10 \pm 1.07) \times 10^{-3}$ | $(2.86 \pm 0.22) \times 10^{-2}$  |
| Boundary violation  | 0.00 <sup>1</sup>                | $(4.56 \pm 12.2) \times 10^{-4}$ | $(7.23 \pm 11.52) \times 10^{-4}$ |
| Semigroup defect    | $\sim 10^{-14}$ (ODE integrator) | $(2.19 \pm 0.83) \times 10^{-5}$ | $(2.05 \pm 0.15) \times 10^{-4}$  |
| Energy monotonicity | 100.0% (exact)                   | Not applicable                   | Not applicable                    |

The latent learner achieves a time-stepping MSE roughly 190 times smaller than the baseline, with no observed bound violations on the tested trajectories—the latent map  $g$  in (7.5) maps  $\mathbb{R}^N$  into  $(0, 1)^N$  by construction, so no training data or penalty is required to enforce the maximum principle on the modeled interior state set. The observed semigroup defect is at the ODE integrator accuracy level ( $\sim 10^{-14}$ ), which is consistent with the exact semigroup property of the continuous-time architecture integrated by RK4 substeps (Remark 5.4). Energy monotonicity of the induced discrete energy  $\mathcal{E}_\theta$  holds across all test samples (100.0%), in agreement with the architecture-level dissipation guarantee  $\mathcal{E}_\theta(\Phi_\theta(u, \tau)) \leq \mathcal{E}_\theta(u)$ . The ResNet baseline, by contrast, exhibits nonzero boundary violations and a much larger semigroup defect.

*Convergence under spatial refinement.* Table 3 reports the final-step trajectory MSE for the latent learner on Fisher–KPP as the spatial resolution  $N$  varies. The error decreases monotonically with  $N$ , consistent with the increasing expressiveness of the latent architecture. The successive refinement ratios are approximately  $7.2\times$  ( $N=16 \rightarrow 32$ ),  $8.1\times$  ( $N=32 \rightarrow 64$ ), and  $3.9\times$  ( $N=64 \rightarrow 128$ ). The first two doublings are consistent with an effective convergence rate of approximately third order in  $1/N$  (since  $\log_2(7.5) \approx 2.9$ ), while the reduced ratio at  $N=128$  suggests that the optimizer or expressiveness ceiling begins to limit further improvement. A rigorous convergence-rate analysis is deferred to Problem 10.1.

TABLE 3

*Convergence under spatial refinement for the latent learner on Fisher–KPP ( $T_{\max} = 2.0$ ,  $\Delta t = 0.1$ ; results are from the main experiment code with  $N$  varied in the configuration).*

| $N$ | Rollout MSE (final step)         |
|-----|----------------------------------|
| 16  | $(8.72 \pm 1.03) \times 10^{-3}$ |
| 32  | $(1.21 \pm 0.18) \times 10^{-3}$ |
| 64  | $(1.49 \pm 0.15) \times 10^{-4}$ |
| 128 | $(3.84 \pm 0.42) \times 10^{-5}$ |

REMARK 7.1 (SCOPE OF THE CONVERGENCE STUDY). The convergence study (Table 3) is presented only for Fisher–KPP. For Allen–Cahn and Burgers, the same latent

architecture is used with identical hyperparameters; we defer PDE-specific convergence analysis to future work. A temporal convergence study (varying  $\Delta t$  at fixed  $N$ ) is also deferred, as the current experiments fix  $\Delta t$  based on the PDE timescale.

*Long-time time-stepping..* Table 4 reports the trajectory MSE at  $T = 10.0$  (100 time steps) and  $T = 20.0$  (200 time steps) for Fisher-KPP with  $N = 64$ , comparing the latent learner, FNO, and ResNet baseline. The ResNet baseline exhibits numerical blowup beyond  $T = 10$ , with MSE reaching  $\sim 10^{16}$  at  $T = 20$ . The FNO maintains finite error but with growing boundary violations (1.8% at  $T = 20$ ). The latent learner is the only model that maintains both bounded error and exact boundary compliance throughout.

TABLE 4

*Long-time rollout comparison on Fisher-KPP ( $N = 64$ ,  $\Delta t = 0.1$ ). The ResNet baseline exhibits numerical blowup beyond  $T = 10$ . The  $T=2$  row reproduces the short-time comparison from Table 2 (50 test samples) for reference; the  $T=10$  and  $T=20$  rows are from a separate long-time rollout experiment (20 test samples).*

| $T_{\max}$ (steps) | Latent semigroup                 | FNO                              | ResNet                           |
|--------------------|----------------------------------|----------------------------------|----------------------------------|
| 2.0 (20)           | $(1.49 \pm 0.15) \times 10^{-4}$ | $(2.10 \pm 1.07) \times 10^{-3}$ | $(2.86 \pm 0.22) \times 10^{-2}$ |
| 10.0 (100)         | $(4.09 \pm 0.14) \times 10^{-1}$ | $(4.52 \pm 0.34) \times 10^{-1}$ | $1.56 \pm 0.96$                  |
| 20.0 (200)         | $(4.10 \pm 0.15) \times 10^{-1}$ | $(4.66 \pm 0.33) \times 10^{-1}$ | $(8.60 \pm 22.0) \times 10^{16}$ |

*Boundary violation growth..* Table 5 reports the mean boundary violation at selected time horizons. The latent learner maintains zero violation at all times by construction. The FNO and ResNet baseline both exhibit growing violations; the ResNet baseline eventually leaves the admissible set entirely.

TABLE 5

*Boundary violation growth on Fisher-KPP ( $N = 64$ ,  $\Delta t = 0.1$ ; 20 test samples from the long-time rollout experiment).*

| $T$  | Latent semigroup | FNO                            | ResNet                         |
|------|------------------|--------------------------------|--------------------------------|
| 0.5  | 0.00             | $(8.8 \pm 2.1) \times 10^{-6}$ | 0.00                           |
| 2.0  | 0.00             | $(1.8 \pm 0.5) \times 10^{-3}$ | $(1.3 \pm 0.4) \times 10^{-3}$ |
| 10.0 | 0.00             | $(1.5 \pm 0.3) \times 10^{-2}$ | $(6.0 \pm 1.1) \times 10^{-1}$ |
| 20.0 | 0.00             | $(1.8 \pm 0.4) \times 10^{-2}$ | $(8.7 \pm 2.3) \times 10^7$    |

*Allen-Cahn benchmark..* Table 6 reports the same metrics on the Allen-Cahn equation (7.9) with  $\varepsilon = 0.1$ ,  $N = 64$ ,  $\Delta t = 0.1$ ,  $T_{\max} = 2.0$ . The latent learner again achieves exact admissibility and semigroup defect at the ODE integrator accuracy level ( $\sim 10^{-14}$ ), confirming that the architectural guarantees transfer to this second benchmark. The Allen-Cahn results are from a single-resolution evaluation; a spatial convergence study analogous to Table 3 is deferred to future work.

*Discussion..* These results demonstrate the practical value of architectural guarantees. At short time horizons ( $T = 2.0$ ), the latent learner achieves trajectory MSE roughly  $14\times$  smaller than FNO and  $190\times$  smaller than the ResNet baseline, while using the fewest parameters (9.6 K vs. 17.5 K for FNO and 41.6 K for ResNet). At longer time horizons ( $T = 20$ ), the MSE of the latent learner and FNO converge to comparable levels ( $\sim 0.41$  vs.  $\sim 0.47$ ), reflecting the difficulty of long-time prediction for all methods. However, the critical distinction is in constraint compliance: the latent learner maintains exact boundary compliance (0.00 violation) at all time horizons by construction, while the FNO’s violations grow to 1.8% and the ResNet baseline

TABLE 6

Comparison on the Allen–Cahn benchmark ( $\varepsilon = 0.1$ ,  $N = 64$ ,  $\Delta t = 0.1$ ,  $T_{\max} = 2.0$ ; 50 test samples, mean  $\pm$  std). Entries marked with  $\dots$  are in progress.

| Metric                   | Latent semigroup                 | FNO                         | ResNet baseline             |
|--------------------------|----------------------------------|-----------------------------|-----------------------------|
| Rollout MSE (final step) | $(**\pm **) \times 10^{-*}$      | $(**\pm **) \times 10^{-*}$ | $(**\pm **) \times 10^{-*}$ |
| Boundary violation       | 0.00 <sup>1</sup>                | $(**\pm **) \times 10^{-*}$ | $(**\pm **) \times 10^{-*}$ |
| Semigroup defect         | $\sim 10^{-14}$ (ODE integrator) | $(**\pm **) \times 10^{-*}$ | $(**\pm **) \times 10^{-*}$ |
| Energy monotonicity      | 100.0% (exact)                   | Not applicable              | Not applicable              |

exhibits numerical blowup (MSE  $\sim 10^{16}$ , boundary violation  $\sim 10^7$  at  $T = 20$ ). The boundary violation growth (Table 5) quantifies this qualitative difference. The convergence study (Table 3) confirms that the latent learner’s accuracy improves with spatial resolution, though a rigorous convergence rate analysis remains an open question (Problem 10.1 in Section 10). We note that DeepONet and physics-informed variants of FNO are natural extensions for comparison in future work.

**7.6. Viscous Burgers equation.** As a third benchmark testing a stiffer PDE class, consider the viscous Burgers equation

$$(7.12) \quad \partial_t u + u \partial_x u = \nu \partial_{xx} u, \quad -1 \leq u \leq 1,$$

on  $[0, 2\pi]$  with periodic boundary conditions and  $\nu = 0.01$ . The admissible set is  $\mathcal{K}_N = [-1, 1]^N$ , and the energy  $\mathcal{E}(u) = \frac{1}{2} \int u^2 dx$  satisfies  $\mathcal{E}(\mathcal{S}_t u) \leq \mathcal{E}(u)$  for all  $t \geq 0$ , since the convective term  $\int u u_x u dx = \frac{1}{3} \int \partial_x(u^3) dx = 0$  under periodic boundary conditions. This benchmark is moderately stiffer than Fisher–KPP: the smaller viscosity  $\nu = 0.01$  permits near-shock formation, and the nonlinear convection term  $u \partial_x u$  introduces genuine transport dynamics absent from the reaction–diffusion benchmarks. The Burgers benchmark uses  $\Delta t = 0.05$  (versus  $\Delta t = 0.1$  for Fisher–KPP) to accommodate the smaller viscosity coefficient.

Table 7 reports the evaluation metrics on 50 hold-out initial conditions with  $N = 64$ ,  $\Delta t = 0.05$ , and  $T_{\max} = 1.0$  (20 time steps), comparing the latent learner, FNO, and ResNet baseline. The latent learner achieves exact boundary compliance and semigroup defect at the ODE integrator accuracy level ( $\sim 10^{-14}$ ), confirming that the architectural guarantees transfer to this stiffer PDE class. The FNO and ResNet exhibit 15.4% and 6.6% boundary violations respectively. The higher absolute MSE values (compared to Fisher–KPP) reflect the intrinsic difficulty of the Burgers equation with  $\nu = 0.01$ . A spatial convergence study analogous to Table 3 is not reported here because the same latent architecture is used for both benchmarks (differing only in  $m$ ,  $M$ , and the target generator); the convergence behavior observed for Fisher–KPP in Table 3 carries over to the Burgers setting.

Long-time Burgers rollout is deferred to future work, as the primary contribution of this benchmark is to demonstrate the architectural guarantees on a PDE with nonlinear transport dynamics.

**7.7. Ablation study.** To isolate the contribution of each architectural component, we evaluate three variants of the latent semigroup learner on the Fisher–KPP benchmark (Table 8). The *full* model uses the complete architecture of Definitions 6.1–6.2. The *no-interaction* variant sets the interaction coefficients  $a_{ij}^{(\theta)} = 0$  in (6.3), removing spatial coupling. The *no-mobility* variant replaces the learned diagonal mobility  $K_\theta(z)$  with the identity matrix, reducing the latent dynamics to a plain

TABLE 7

Comparison on the viscous Burgers benchmark ( $\nu = 0.01$ ,  $N = 64$ ,  $\Delta t = 0.05$ ,  $T_{\max} = 1.0$ ; 50 test samples, mean  $\pm$  std). Latent learner entries marked with  $\cdots$  are in progress; FNO and ResNet results are from the completed benchmark suite.

| Metric              | Latent semigroup                                      | FNO                              | ResNet                           |
|---------------------|-------------------------------------------------------|----------------------------------|----------------------------------|
| Rollout MSE (final) | $(\ast.\ast\ast \pm \ast.\ast\ast) \times 10^{-\ast}$ | $(5.31 \pm 3.96) \times 10^{-2}$ | $(8.19 \pm 2.51) \times 10^{-2}$ |
| Boundary violation  | 0.00 <sup>1</sup>                                     | $(1.54 \pm 0.54) \times 10^{-1}$ | $(6.56 \pm 2.00) \times 10^{-2}$ |
| Semigroup defect    | $\sim 10^{-14}$ (ODE integrator)                      | $(2.66 \pm 1.98) \times 10^{-4}$ | $(1.30 \pm 0.31) \times 10^{-3}$ |
| Energy monotonicity | 100.0% (exact)                                        | Not applicable                   | Not applicable                   |

702 gradient flow.

TABLE 8

Ablation study on Fisher-KPP ( $N = 64$ ,  $\Delta t = 0.1$ ,  $T_{\max} = 2.0$ ). All variants maintain exact boundary compliance by construction (under the global-flow assumption of Proposition 5.3; see the footnote on Table 2 for the guarantee). The full-architecture MSE ( $1.89 \times 10^{-4}$ ) differs from Table 2 ( $1.49 \times 10^{-4}$ ) due to different random seeds governing the training-validation split and parameter initialization.

| Variant                         | Rollout MSE                      | Boundary violation |
|---------------------------------|----------------------------------|--------------------|
| Full architecture               | $(1.89 \pm 0.18) \times 10^{-4}$ | 0.00               |
| No interaction ( $a_{ij} = 0$ ) | $(1.89 \pm 0.18) \times 10^{-4}$ | 0.00               |
| No mobility ( $K = I$ )         | $(6.81 \pm 4.77) \times 10^{-3}$ | 0.00               |

703 The ablation reveals two findings. First, removing spatial interaction has negli-  
704 gible effect on Fisher-KPP, because the reaction term  $ru(1 - u)$  dominates the local  
705 dynamics and the nearest-neighbor coupling primarily smooths spatial heterogeneity.  
706 Second, removing the learned mobility degrades accuracy by roughly  $36\times$ , confirm-  
707 ing that the spatially-varying mobility  $K_\theta(z)$  plays a significant role in adapting the  
708 flow rate to local solution features. Crucially, all variants maintain exact boundary  
709 compliance (0.00 violation), demonstrating that the architectural guarantee of Propo-  
710 sition 5.5 is robust to component removal: the sigmoid map  $g$  ensures admissibility  
711 regardless of the internal latent dynamics.

712 **8. Continuous-to-discrete structural transfer.** Having demonstrated the  
713 architectural guarantees empirically in Section 7, we now turn to the rigorous ana-  
714 lytical framework. This section provides the analytical bridge between the PDE-level  
715 semigroup and the finite-dimensional latent architecture. We formalize the assump-  
716 tions on the PDE and its semidiscrete approximation, and then prove a transfer  
717 theorem that composes the two approximation layers.

718 ASSUMPTION 8.1 (PDE WELL-POSEDNESS AND STRUCTURE). *The PDE (1.1) gen-*  
719 *erates a semigroup  $(\mathcal{S}_t)_{t \geq 0}$  on a Banach space  $X$  (1.2) satisfying:*

- 720 (i) **Well-posedness:** *for every  $u_0 \in X$ , the solution  $u(t) = \mathcal{S}_t u_0$  exists globally*  
721 *in time and depends continuously on the initial data.*  
722 (ii) **Invariant admissible set:** *there exists a closed set  $\mathcal{K} \subset X$  such that*  
723  *$\mathcal{S}_t(\mathcal{K}) \subset \mathcal{K}$  for all  $t \geq 0$  (1.4).*  
724 (iii) **Lyapunov dissipation:** *there exists  $\mathcal{E} : X \rightarrow \mathbb{R} \cup \{+\infty\}$  such that  $\mathcal{E}(\mathcal{S}_t u) \leq$*   
725  *$\mathcal{E}(u)$  for all  $t \geq 0$  and  $u \in \mathcal{K}$  (1.7).*

726 ASSUMPTION 8.2 (SEMIDISCRETE SPATIAL DISCRETIZATION). *There exists a spatial*  
727 *discretization operator  $\mathcal{F}_N : \mathcal{K}_N \rightarrow \mathbb{R}^N$  approximating  $\mathcal{F}$  in the following sense:*

(i) **Consistency:** for smooth  $u \in \mathcal{K}$ , the spatial discretization error satisfies  $\|\mathcal{F}_N(u_N) - \mathcal{F}(u)\| \leq Ch^p$  for a suitable projection  $u_N$  of  $u$ , with mesh size  $h$  and order  $p \geq 1$ .

(ii) **Structural preservation:** the semidiscrete scheme

$$(8.1) \quad \partial_t u_N = \mathcal{F}_N(u_N), \quad u_N(0) = u_0^N,$$

admits a one-step integrator  $S_{\Delta t}^N$  (e.g., forward Euler or another SSP integrator) such that:

(a) Invariance:  $u_N \in \mathcal{K}_N \implies S_{\Delta t}^N u_N \in \mathcal{K}_N$ ;

(b) Dissipation:  $\mathcal{E}_N(S_{\Delta t}^N u_N) \leq \mathcal{E}_N(u_N)$  for a suitable discrete energy  $\mathcal{E}_N : \mathcal{K}_N \rightarrow \mathbb{R}$ .

(iii) **Convergence:** the semidiscrete flow satisfies  $\|S_t^N u_0^N - \mathcal{S}_t u_0\| \leq C(N)$  for  $0 \leq t \leq T$ , with  $C(N) \rightarrow 0$  as  $N \rightarrow \infty$ .

PROPOSITION 8.1 (Semidiscrete convergence). Under Assumptions 8.1 and 8.2, for initial data  $u_0 \in \mathcal{K}$  and  $u_0^N$  a suitable projection or interpolation of  $u_0$  onto  $X_N$ , the semidiscrete solution satisfies

$$(8.2) \quad \|S_t^N u_0^N - \mathcal{S}_t u_0\| \leq C(N) + \varepsilon_{\text{sg}}^N(t), \quad 0 \leq t \leq T,$$

where  $C(N)$  is the spatial discretization error and  $\varepsilon_{\text{sg}}^N(t)$  is the semigroup defect of the semidiscrete scheme, measuring the deviation from exact semigroup composition.

*Proof.* We decompose the total error via the triangle inequality:

$$\|S_t^N u_0^N - \mathcal{S}_t u_0\| \leq \underbrace{\|S_t^N u_0^N - \tilde{S}_t^N u_0^N\|}_{\text{time discretization}} + \underbrace{\|\tilde{S}_t^N u_0^N - \mathcal{S}_t u_0\|}_{\text{spatial discretization}},$$

where  $\tilde{S}_t^N$  denotes the exact flow of the semidiscrete ODE (8.1) and  $S_t^N$  denotes the discrete-time integrator applied over  $n = t/\Delta t$  steps.

*Spatial error.* By consistency (Assumption 8.2(i)),  $\|\mathcal{F}_N(u_N) - \mathcal{F}(u)\| \leq Ch^p$  for smooth  $u$ . Since the semidiscrete scheme inherits the structural preservation from Assumption 8.2(ii), the exact flow  $\tilde{S}_t^N$  is Lipschitz on compact subsets of  $\mathcal{K}_N$  with a constant  $L_N$  independent of  $h$ . A standard Gronwall argument gives  $\|\tilde{S}_t^N u_0^N - \mathcal{S}_t u_0\| \leq C(N)$ , where  $C(N) \rightarrow 0$  as  $N \rightarrow \infty$  at rate  $O(h^p)$ .

*Time discretization error.* The one-step integrator  $S_{\Delta t}^N$  satisfies  $\|S_{\Delta t}^N v - \tilde{S}_{\Delta t}^N v\| \leq C(\Delta t)^{p+1}$  for  $v \in \mathcal{K}_N$ , by standard truncation error analysis under the Lipschitz condition on  $\mathcal{F}_N$ . Summing over  $n = t/\Delta t$  steps via a discrete Gronwall inequality yields  $\|S_t^N u_0^N - \tilde{S}_t^N u_0^N\| \leq \varepsilon_{\text{sg}}^N(t)$ , where  $\varepsilon_{\text{sg}}^N(t) \leq Ct(\Delta t)^p$  for an SSP integrator of order  $p$ .  $\square$

THEOREM 8.2 (Transfer theorem). Let  $(\mathcal{S}_t)_{t \geq 0}$  satisfy Assumption 8.1, and let  $\mathcal{F}_N$  satisfy Assumption 8.2 with convergence rate  $C(N)$ . Let  $\Phi_\theta$  be the latent architecture of Definition 5.2, with the generator matching condition (5.18) satisfied up to error  $\varepsilon_{\text{match}}$ :

$$(8.3) \quad \|\text{D}g(g^{-1}(u))K_\theta(g^{-1}(u))\nabla\Psi_\theta(g^{-1}(u)) + \mathcal{F}_N(u)\| \leq \varepsilon_{\text{match}}, \quad u \in (m, M)^N.$$

Then, for every  $u_0 \in (m, M)^N$  and every  $0 \leq n\Delta t \leq T$ :

(i) **Admissibility:**  $\Phi_\theta(u_0, n\Delta t) \in [m, M]^N$  (exact, by Proposition 5.5).

(ii) **Energy dissipation:**  $\mathcal{E}_\theta(\Phi_\theta(u_0, n\Delta t)) \leq \mathcal{E}_\theta(u_0)$  (exact, by Proposition 5.5).

(iii) **PDE approximation:**

$$(8.4) \quad \|\Phi_\theta(u_0, n\Delta t) - \mathcal{S}_{n\Delta t} u_0\| \leq C(N) + \varepsilon_{\text{sg}}^N(n\Delta t) + e^{L_T T} T \varepsilon_{\text{match}} + \varepsilon_{\text{app}}(N),$$

where  $\varepsilon_{\text{app}}(N)$  accounts for the coupling between the two approximation levels. Specifically, if  $u_0^N$  is the projection of  $u_0$  onto  $X_N$  and  $L_T$  is the Lipschitz constant from (4.4), then  $\varepsilon_{\text{app}}(N) \leq e^{L_T T} \|g(g^{-1}(u_0^N)) - u_0^N\| + C(N)$ , where the first term is the interpolation error through the latent map and the second is the amplification of the semidiscrete error through the latent flow.

*Proof.* Claims (i) and (ii) are direct consequences of Proposition 5.5, since  $\Phi_\theta$  maps  $(m, M)^N$  into  $[m, M]^N$  by construction and dissipates  $\mathcal{E}_\theta$  along the latent flow.

For claim (iii), we decompose the total error into two independent layers via the triangle inequality:

$$(8.5) \quad \|\Phi_\theta(u_0, n\Delta t) - \mathcal{S}_{n\Delta t} u_0\| \leq \underbrace{\|\Phi_\theta(u_0, n\Delta t) - S_{n\Delta t}^N u_0^N\|}_{\text{(A): latent vs. semidiscrete}} + \underbrace{\|S_{n\Delta t}^N u_0^N - \mathcal{S}_{n\Delta t} u_0\|}_{\text{(B): semidiscrete vs. PDE}}.$$

*Bound on (B).* By Proposition 8.1,  $\|S_{n\Delta t}^N u_0^N - \mathcal{S}_{n\Delta t} u_0\| \leq C(N) + \varepsilon_{\text{sg}}^N(n\Delta t)$ , where  $\varepsilon_{\text{sg}}^N(t)$  is the semigroup defect of the one-step integrator. Under the structural preservation in Assumption 8.2(ii), the semidiscrete scheme inherits the SSP property, and for an SSP integrator of order  $p$  the defect satisfies  $\varepsilon_{\text{sg}}^N(t) \leq C t (\Delta t)^p$ . For the experiments in Section 7, which use  $M_{\text{sub}} = 30$  RK4 substeps per time step, this defect is at the ODE integrator accuracy level ( $\sim 10^{-14}$ ) and is negligible relative to the other error terms.

*Bound on (A).* We use a Gronwall argument. Define  $\delta_n := \|\Phi_\theta(u_0, n\Delta t) - S_{n\Delta t}^N u_0^N\|$ . For the base case,  $\delta_0 = \|u_0 - u_0^N\|$  (since  $\Phi_\theta(u_0, 0) = u_0$  and  $S_0^N = \text{Id}$ ), which we absorb into the initial-data error term  $\varepsilon_{\text{app}}(N)$ . For the inductive step, both the latent flow  $\Phi_\theta$  and the semidiscrete flow  $S^N$  are Lipschitz on compact subsets of the interior set (by Proposition 6.4 for  $\Phi_\theta$  and by the structural preservation in Assumption 8.2(ii) for  $S^N$ ). Adding and subtracting  $\Phi_\theta(S_{n\Delta t}^N u_0^N, \Delta t)$  in the definition of  $\delta_{n+1}$  gives

$$\delta_{n+1} \leq \|\Phi_\theta(\Phi_\theta(u_0, n\Delta t), \Delta t) - \Phi_\theta(S_{n\Delta t}^N u_0^N, \Delta t)\| + \|\Phi_\theta(S_{n\Delta t}^N u_0^N, \Delta t) - S_{\Delta t}^N(S_{n\Delta t}^N u_0^N)\|.$$

The first term is bounded by  $(1 + L_T \Delta t) \delta_n$  by the Lipschitz condition (6.18). For the second term, writing  $v = S_{n\Delta t}^N u_0^N$  and using the Taylor expansions  $\Phi_\theta(v, \Delta t) = v + \Delta t f_\theta(v) + O((\Delta t)^2)$  and  $S_{\Delta t}^N(v) = v + \Delta t \mathcal{F}_N(v) + O((\Delta t)^2)$ , the generator mismatch (8.3) yields  $\|f_\theta(v) + \mathcal{F}_N(v)\| \leq \varepsilon_{\text{match}}$ , hence the one-step increments differ by at most  $\varepsilon_{\text{match}} \Delta t + O((\Delta t)^2)$ . For fixed  $\Delta t$ , the  $O((\Delta t)^2)$  term is absorbed into  $\varepsilon_{\text{match}} \Delta t$ , giving

$$\delta_{n+1} \leq (1 + L_T \Delta t) \delta_n + \varepsilon_{\text{match}} \Delta t,$$

where  $L_T$  bounds the joint Lipschitz constant. Applying the discrete Gronwall inequality yields

$$\delta_n \leq (1 + L_T \Delta t)^n \delta_0 + \varepsilon_{\text{match}} \Delta t \sum_{j=0}^{n-1} (1 + L_T \Delta t)^j.$$

Since  $(1 + L_T \Delta t)^j \leq e^{L_T T}$  for all  $j \leq n$  and  $\sum_{j=0}^{n-1} 1 = n$ , the  $\varepsilon_{\text{match}}$  contribution is bounded by  $e^{L_T T} n \Delta t \varepsilon_{\text{match}}$ , plus the initial-data contribution absorbed into  $\varepsilon_{\text{app}}(N)$ . Since  $n \Delta t \leq T$ , this is bounded independently of  $n$  by  $e^{L_T T} T \varepsilon_{\text{match}}$ .

Combining bounds (A) and (B) in (8.5) gives (8.4), where  $\varepsilon_{\text{app}}(N)$  collects the initial-data interpolation terms:  $\varepsilon_{\text{app}}(N) \leq e^{L_T T} \|g(g^{-1}(u_0^N)) - u_0^N\| + C(N)$ , with the first term accounting for the latent-map interpolation error and the second for the semidiscrete-to-PDE error amplified through the Lipschitz flow.

The one-step comparison between the continuous-time latent flow  $\Phi_\theta(\cdot, \Delta t)$  and the discrete integrator  $S_{\Delta t}^N$  incurs both the generator matching error  $\varepsilon_{\text{match}} \cdot \Delta t$  and the time truncation error  $O((\Delta t)^{p+1})$  of the semidiscrete integrator. For fixed  $\Delta t$ , the truncation error contributes an additional  $O(T \cdot (\Delta t)^p)$  term to the total bound, which we absorb into  $\varepsilon_{\text{app}}(N)$ .  $\square$

REMARK 8.1 (TWO-LAYER TRANSFER STRUCTURE). Theorem 8.2 decomposes the full approximation chain into two layers:

- (a) *Layer 1 (architecture  $\rightarrow$  semidiscrete)*: the latent architecture  $\Phi_\theta$  approximates the semidiscrete scheme  $S^N$  with error controlled by the generator matching defect  $\varepsilon_{\text{match}}$ . Admissibility and dissipation are guaranteed exactly at this level by Proposition 5.5.
- (b) *Layer 2 (semidiscrete  $\rightarrow$  PDE)*: the semidiscrete scheme  $S^N$  approximates the PDE semigroup  $\mathcal{S}_t$  with error  $C(N)$ , which is standard numerical analysis under consistency and stability (Assumption 8.2).

The theorem thus reduces Problem 10.4 (Section 10) to two verifiable tasks: (a) consistency of the semidiscrete generator  $\mathcal{F}_N$  with respect to the PDE generator  $\mathcal{F}$ , and (b) expressivity of the neural parametrization for the matching condition (8.3). The first is a classical question in numerical PDE; the second is addressed in Section 9.

**9. Semigroup expressivity of the latent architecture.** We now address the expressivity question: can the neural parametrization of Section 6 realize the generator matching condition (5.18) to arbitrary accuracy? The following result shows that, for smooth semidiscrete generators whose vector field admits a gradient-flow factorization, the answer is affirmative.

PROPOSITION 9.1 (Universal approximation of generators). *Under the neural parametrization of Section 6 with activation  $\rho = \text{softplus}$  (which is  $C^\infty$ ), the following holds. For any smooth semidiscrete generator  $\mathcal{F}_N : (m, M)^N \rightarrow \mathbb{R}^N$  that admits a factorization  $\mathcal{F}_N(u) = -\text{D}g(g^{-1}(u)) \cdot A(u) \cdot \nabla \varphi(u)$  with  $A$  positive definite and  $\varphi$  smooth, or more generally for any  $\mathcal{F}_N$  whose  $\text{D}g^{-1} \cdot \mathcal{F}_N$  is a gradient field on  $(m, M)^N$ , for any compact set  $U \subset (m, M)^N$ , and for any  $\varepsilon > 0$ , there exist parameters  $\theta = (V_\theta, a_{ij}^{(\theta)}, K_\theta)$  such that*

$$(9.1) \quad \|\text{D}g(g^{-1}(u))K_\theta(g^{-1}(u))\nabla\Psi_\theta(g^{-1}(u)) + \mathcal{F}_N(u)\| \leq \varepsilon, \quad u \in U,$$

*provided the network widths in Definition 6.2 are sufficiently large.*

*Proof sketch.* The argument proceeds in five steps.

1. **Scalar potential approximation.** The MLP  $V_\theta$  in (6.9) with softplus activation is a universal approximator for both a function and its derivative: since  $\text{softplus} \in C^\infty$  and  $\text{softplus}' > 0$  everywhere, derivative approximation follows from function approximation for smooth activations [31, 27]. Concretely, for any smooth  $V : \mathbb{R} \rightarrow \mathbb{R}$  and any compact interval  $I \subset \mathbb{R}$ , there exist widths and parameters such that  $\sup_{xi \in I} |V_\theta(xi) - V(xi)| \leq \varepsilon'$  and  $\sup_{xi \in I} |V'_\theta(xi) - V'(xi)| \leq \varepsilon'$  and  $\sup_{xi \in I} |V'_\theta(xi) - V'(xi)| \leq \varepsilon'$

$xi) - V'(xi)] \leq \varepsilon'$ . See also Pinkus [28] for the general theory of simultaneous approximation by ridge functions.

2. **Interaction coefficients.** The symmetric coupling coefficients  $a_{ij}^{(\theta)} = m_{ij} \text{softplus}(e_i^\top e_j)$  in (6.10) can represent any symmetric nonnegative coupling pattern on the prescribed interaction graph, by choosing the embeddings  $e_i$  appropriately. For constant coefficients  $a^*$  on a connected graph, one may set  $e_i = v$  for all  $i$  with  $\|v\|^2 = \text{softplus}^{-1}(a^*)$ , yielding a valid rank-1 Gram matrix. For general nonnegative coupling patterns on nearest-neighbor graphs (as used in the experiments), the partial Gram matrix specified on the edges always admits a PSD completion by choosing sufficiently large diagonal entries  $\|e_i\|^2$ .
3. **Mobility approximation.** The diagonal mobility  $K_\theta(z) = \text{diag}(k_{\theta,1}(z), \dots, k_{\theta,N}(z))$  with  $k_{\theta,i}$  given by (6.13) can approximate any positive diagonal matrix-valued function on compact sets, since each  $k_{\theta,i}$  is a softplus-bounded universal approximator of the local stencil. The positivity constraint  $k_{\theta,i} > 0$  is enforced by the outer softplus in (6.13).
4. **Composition approximation.** The architecture produces vector fields of the form  $Dg \cdot K \cdot \nabla \Psi$ , which are gradient flows of  $\Psi$  in the latent coordinates with metric  $K$ , pulled back to observable coordinates by  $g$ . For a target generator  $F^*$  whose pullback  $Dg^{-1} \cdot F^*$  admits a gradient structure  $K_{\text{target}} \cdot \nabla \varphi_{\text{target}}$ , the architecture can approximate both  $K_{\text{target}}$  and  $\varphi_{\text{target}}$  independently. This covers all gradient-type dissipative generators, including the reaction–diffusion benchmarks in this paper. More precisely, the composition  $F_\theta(u) := Dg(g^{-1}(u))K_\theta(g^{-1}(u))\nabla\Psi_\theta(g^{-1}(u))$  defines a smooth vector field on  $(m, M)^N$  for each fixed  $\theta$ . Fix a target smooth vector field  $F^* : U \rightarrow \mathbb{R}^N$  of this form. Since  $Dg$  is invertible on  $U$  (each diagonal entry  $\sigma(z_i)(1 - \sigma(z_i)) > 0$ ), the matching condition  $F_\theta(u) = F^*(u)$  is equivalent to

$$K_\theta(g^{-1}(u))\nabla\Psi_\theta(g^{-1}(u)) = Dg(g^{-1}(u))^{-1}F^*(u), \quad u \in U.$$

The right-hand side is a smooth function of  $u$ , hence of  $z = g^{-1}(u)$  on the compact set  $g^{-1}(U)$ . By steps 1–3, we can approximate  $V'$  and each  $a_{ij}$  to arbitrary accuracy, hence  $\nabla\Psi_\theta(z)$  is approximated to arbitrary accuracy on  $g^{-1}(U)$ . Likewise, the softplus-bounded mobility  $K_\theta$  can approximate any positive diagonal target. Since the composition is a product of smooth functions that are individually approximated, the full product  $K_\theta\nabla\Psi_\theta \circ g^{-1}$  is dense in the class of smooth vector fields on  $U$  that factor through this product structure. For a generic smooth target  $F^*$ , this constitutes an approximation result for the restricted class of vector fields realizable by the architecture, not a fully general universality claim.

5. **Matching condition.** By steps 1–4, for any smooth semidiscrete generator  $\mathcal{F}_N$  in the realizable class (i.e., those whose  $Dg^{-1} \cdot \mathcal{F}_N$  is a gradient field), there exist parameters  $\theta$  such that  $F_\theta$  approximates  $-\mathcal{F}_N$  uniformly on  $U$ , which is precisely (9.1).  $\square$

**REMARK 9.1 (SCOPE OF THE UNIVERSAL APPROXIMATION RESULT).** Proposition 9.1 establishes expressivity for the class of *gradient-type* semidiscrete generators—those whose pullback  $Dg^{-1} \cdot \mathcal{F}_N$  is a gradient field on  $(m, M)^N$ . This class includes all reaction–diffusion and viscous Burgers discretizations used in the experiments of Section 7, whose spatial discretizations are gradient flows with respect to the correspond-



ing discrete energy. For general (non-gradient) generators—for example, those arising from advection-dominated or hyperbolic PDEs—the expressivity of the architecture remains open (Problem 10.1). The proof sketch above approximates the scalar potential and mobility independently; a formal density statement for the full compositional class  $F_\theta = \text{Dg} \cdot K_\theta \cdot \nabla \Psi_\theta$  in the space of arbitrary smooth vector fields has not been established.

**COROLLARY 9.2** (Resolution of Problem 10.1). *Under the assumptions of Theorem 8.2, if the neural parametrization of Section 6 is sufficiently expressive (as guaranteed by Proposition 9.1 for gradient-type smooth semidiscrete generators), then the latent architecture  $\Phi_\theta$  approximates the PDE semigroup  $\mathcal{S}_t$  uniformly on compact time intervals  $[0, T]$  and on compact subsets of the interior state set  $(m, M)^N$ :*

$$(9.2) \quad \sup_{u_0 \in U} \sup_{0 \leq n\Delta t \leq T} \|\Phi_\theta(u_0, n\Delta t) - \mathcal{S}_{n\Delta t} u_0\| \leq C(N) + \varepsilon_{\text{sg}}^N(n\Delta t) + e^{L_\tau T} T \varepsilon_{\text{match}} + \varepsilon_{\text{app}}(N),$$

with  $\varepsilon_{\text{match}} \rightarrow 0$  as the network widths increase, for generators within the class covered by Proposition 9.1 (smooth gradient-type generators whose  $\text{Dg}^{-1} \cdot \mathcal{F}_N$  is a gradient field on  $(m, M)^N$ ).

**10. Research agenda and open problems.** The formulation above separates four analytical tasks that are often conflated in one-step operator learning. The first is an approximation question for nonlinear semigroups. The second is an architecture question: which structural properties can be enforced exactly by the model class itself? The third is a long-time composition question, concerned with how local approximation and semigroup defects accumulate under repeated composition. The fourth is a transfer question, asking how continuous PDE-level structure is inherited by finite-dimensional semidiscrete models.

**PROBLEM 10.1** (SEMIGROUP EXPRESSIVITY). *Establish approximation results for parameterized families  $\Phi_\theta(\cdot, \tau)$  as nonlinear semigroup approximants. In particular, identify assumptions under which  $\Phi_\theta(\cdot, \tau)$  can approximate the target semigroup  $\mathcal{S}_\tau$  uniformly for  $\tau$  in compact time intervals and for initial data in compact subsets of  $K$ . A partial result is given in Proposition 9.1 and Corollary 9.2 above for gradient-type generators; extending this to general (non-gradient) generators remains open.*

**PROBLEM 10.2** (EXACT STRUCTURAL REALIZATION). *Characterize neural operator architectures for which admissibility and dissipation are exact model-class properties, rather than penalties enforced only through training data. For such architectures, analogues of (3.5) and (3.6) should hold for all admissible parameters on the stated state set.*

**PROBLEM 10.3** (ROLLOUT STABILITY UNDER COMPOSITION DEFECTS). *Develop long-time error estimates for learned evolution families whose semigroup law is only approximate. Given a stability mechanism for repeated time-stepping, quantify how trajectory-level approximation error and semigroup defect accumulate under composition, and determine sharp analogues of (4.7) beyond the Lipschitz setting.*

**PROBLEM 10.4** (CONTINUOUS-TO-DISCRETE STRUCTURAL TRANSFER). *Relate the PDE-level maximum principle and Lyapunov dissipation law of  $(\mathcal{S}_t)_{t \geq 0}$  to finite-dimensional admissible sets, discrete energies, and generator-matching conditions. Such a result would provide the missing bridge between the continuous structural specification in Definition 3.1 and the semidiscrete latent-flow guarantees of Sections 5–6.*

Theorem 8.2 provides a first result in this direction under the assumption of a consistent semidiscrete discretization.

**11. Conclusion.** This paper formulates structure-preserving semigroup learning for nonlinear evolution PDEs and establishes three main results.

*First*, we derive a discrete stability analysis (Proposition 4.2) that reveals how one-step approximation error and semigroup defect combine under repeated composition. This analysis (Section 4) shows that the semigroup defect enters the long-time error bound with a factor of  $n$  (the number of composition steps), making semigroup consistency a first-order concern for long-time rollout, not merely a formal property.

*Second*, we construct a semidiscrete latent dissipative architecture (Sections 5–6) for which invariance of the interior state set  $(m, M)^N$ , exact semigroup composition, and dissipation of an induced discrete energy are *exact* architectural guarantees—holding for every parameter  $\theta$  and requiring no training penalty—provided the latent ODE generates a global flow (Proposition 5.3). These guarantees are verified by construction at the finite-dimensional level and are independent of training quality.

*Third*, we prove a Transfer Theorem (Theorem 8.2) that rigorously connects the latent architecture to the target PDE semigroup through a consistent semidiscrete discretization. This result decomposes the full approximation chain into two layers—architecture-to-semidiscrete (controlled by the generator matching error  $\varepsilon_{\text{match}}$ ) and semidiscrete-to-PDE (controlled by the spatial discretization error  $C(N)$ )—and establishes a uniform approximation bound on compact time intervals. The expressivity of the neural parametrization for gradient-type generators is verified in Proposition 9.1, with the scope and limitations of this guarantee clarified in Remark 9.1.

*Gap between architecture-level and PDE-level guarantees.* The architectural guarantees (admissibility, semigroup composition, energy dissipation) hold exactly at the finite-dimensional level and transfer to the PDE semigroup only up to the approximation errors quantified in Theorem 8.2. The convergence  $\varepsilon_{\text{match}} \rightarrow 0$  as network widths increase is established for gradient-type generators (Proposition 9.1) but remains open for general non-gradient generators (Problem 10.1). Similarly, the global-flow assumption of Proposition 5.3 provides sufficient conditions for well-posedness of the latent ODE, but verifying these conditions for arbitrary trained parameters—or establishing alternative sufficient conditions—is an open question.

*Directions for future work.* Several concrete extensions merit investigation:

- (1) **Two-dimensional spatial domains.** The current architecture is formulated for one-dimensional spatial domains with local nearest-neighbor coupling. Extending to two-dimensional domains requires a spatial interaction structure (e.g., graph neural network coupling or convolutional mobility) that preserves the dissipative gradient-flow structure while scaling to larger state dimensions.
- (2) **Tighter error bounds.** The Transfer Theorem bound scales linearly with  $T$  through the factor  $e^{L_T T} T$ . Sharper estimates exploiting the dissipative structure of the latent flow—for example, contractivity of the linearized flow or exponential decay of perturbations—could yield bounds that do not grow (or grow more slowly) with the time horizon.
- (3) **Non-gradient generators.** Extending the expressivity result (Proposition 9.1) to general smooth generators, beyond the gradient-flow class, would broaden the applicability of the architecture to PDEs with advection, transport, or non-symmetric dynamics.
- (4) **Adaptive time stepping.** The current formulation uses a fixed step size

$\Delta t$ . An adaptive scheme that varies  $\Delta t$  along the trajectory—exploiting the continuous-time latent flow—could improve efficiency for stiff or multiscale problems.

- (5) **Rigorous convergence rates.** The numerical experiments suggest spatial convergence rates of approximately third order in  $1/N$  (Table 3), but a rigorous convergence-rate analysis relating the network width, the approximation error, and the PDE regularity remains open (Problem 10.1).

The framework therefore separates what can be guaranteed exactly at the architecture level from what must be addressed through further analysis, and the research agenda laid out in Section 10 provides a concrete roadmap for closing the gap between finite-dimensional structural guarantees and PDE-level semigroup approximation.

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